American English Intonation

A complete reference for an advanced French speaker

Contents

Part XII — Reference · notation · weak forms · nucleus rules · suffixes · tunes · self-diagnostic · myths · resources · glossary · references

PART 0 — HOW TO USE THIS

0.1 What this is

A reference, not a course. Read it through once slowly, then return to pieces. Parts I–II are concepts. Parts III–VII are the content of English prosody, smallest unit to largest. Part VIII is above the sentence. Parts IX–X are *how to learn*. Part XI is a drill library. Part XII is tables you'll consult constantly.

0.2 Notation

Symbol	Meaning
`CAPS`	the **nucleus** — the one main accent of a thought group
`'`	primary word stress, placed before the syllable (`im'portant`)
`ˌ`	secondary word stress (`ˌuniver'sity`)
`\`	falling pitch movement beginning on that syllable
`/`	rising
`\/`	fall-rise
`/\`	rise-fall
`→`	level / plateau
single pipe	minor thought-group boundary
double pipe	major boundary (sentence or topic end)
`ə`	schwa — the reduced, colourless vowel
`ɚ`	r-coloured schwa, as in *butt**er***
`‿`	linking across a word boundary

Example: `I 'didn't 'say he \/STOLE it` — accents on *didn't* and *say*, nucleus on *stole* carrying a fall-rise. Meaning: "…I said something else."

0.3 Evidence grades

Prosody research is uneven. Some findings are replicated and settled; others rest on a single study, or are actively disputed, or are teaching lore that has never been tested. Every substantive empirical claim below carries a grade so you can tell which is which:

- **[E]** — **Established.** Replicated, broadly agreed.
- **[T]** — **Tendency.** Real and measurable, with substantial variation or exceptions.
- **[C]** — **Contested.** Studies disagree, or the measurement is methodologically hard.

- **[P] — Practitioner heuristic.** Not a research finding. Something teachers and coaches use because it works; treat as a working rule, not a fact.
- **[I] — My inference.** A conclusion I've drawn from combining findings. Reasonable, but mine.

When you find advice elsewhere that contradicts something here, check the grade first. Disagreement with a **[P]** or **[I]** is unremarkable. Disagreement with an **[E]** is worth investigating.

0.4 Do this before reading further: the day-one baseline

Twenty minutes. You will want this recording in six months, and you cannot make it retroactively.

Record four things, in one file, dated:

1. **The diagnostic passage** below, read aloud once, cold, no preparation.
2. **Twenty words** read from a list: *important, development, comfortable, necessary, photography, administration, category, laboratory, employee, determine, image, purchase, separate (adjective), separate (verb), thirteen, thirty, greenhouse, green house, record (noun), record (verb).*
3. **Two minutes of spontaneous speech** on: "explain your job to someone outside your field."
4. **One minute of spontaneous speech** on a topic you pick at random from the news, with no preparation at all.

The diagnostic passage (written to load every French-speaker weak point — cognate stress, `-ary`/`-ory` words, long Latinate words, weak forms, flapping, compounds, long tails after the nucleus, and thought-group boundaries):

> The development team's report was interesting, though not particularly comfortable reading. It's necessary, I think, to separate the temporary problems from the permanent ones. The laboratory results — which nobody had actually analysed — suggested that the equipment was the difficulty, not the management. Our category of customer expects a guarantee, and thirteen of the thirty complaints we received last February were about exactly that. I'm not saying it's impossible. I'm saying we'd have to be a lot more comparable to our competitors before anyone would take the opportunity seriously. Could you check the computer? Is it important?

Then measure four numbers in Praat (see 9.6 for how):

Measure	How	Your day-one value
Strong:weak duration ratio	longest vs shortest syllable in *administration*	
Pitch span	max F0 minus min F0 across the passage, in semitones	
Pitch span in **French**	same measure, reading any French paragraph	
Articulation rate	syllables ÷ (total time − pause time)	

That third row is the one nobody thinks to take, and it's the most diagnostic thing on the list. If your French span is much wider than your English span, your problem is suppression, not capacity — and that is very good news, because suppression is easier to fix than capacity.

Repeat the whole protocol monthly. Never listen to the old recordings for the first three months; then listen to month 1 and be encouraged.

PART I — ORIENTATION

1.1 What "intonation" means

Two senses, and conflating them makes people practise the wrong thing.

Narrow sense (melody): the pattern of pitch movement across an utterance. The tune.

Broad sense (prosody): everything about speech that isn't the identity of individual consonants and vowels — pitch, loudness, duration, rhythm, tempo, pausing, voice quality.

This document uses the broad sense, because that's what you need. The narrow sense is Part VII. The map:

```
PROSODY
├── RHYTHM        which syllables are long/short; where beats fall; compression
├── STRESS        relative prominence of syllables
│   ├── lexical / word stress     a stored property of the word
│   └── prominence / accent       a choice made per utterance
├── INTONATION    the pitch tune: pitch accents + edge tones
├── PHRASING      chunking; boundaries; pauses
├── TEMPO         speech rate and its local variation
└── VOICE QUALITY creak, breathiness, nasality, register, resonance
```

Four physical parameters do all of it: **F0** (heard as pitch), **duration**, **intensity** (loudness), and **spectral quality** (mostly vowel quality). Every prosodic phenomenon in English is some combination of these four. *Which* combination, in what proportion, is where languages differ — and where your accent lives.

1.2 Why prosody outranks segments for your goals

You said: clear first, correct second, less French third. That ordering is well chosen, and it maps onto the evidence.

Prosodic errors damage comprehension more than segmental errors. [T] Several studies found suprasegmental problems more destructive than consonant/vowel problems, and prosody-focused instruction producing larger comprehensibility gains than segment-focused instruction (Anderson-Hsieh et al. 1992; Johansson 1978; Derwing, Munro & Wiebe 1998). A minority of studies found the reverse (Koster & Koet 1993; Fayer & Krasinski 1987). Graded **[T]** rather than **[E]** because the literature genuinely disagrees at the edges — but the weight is clearly on one side.

Misplaced word stress is unusually destructive. [E] Field (2005) shifted stress to the wrong syllable and found intelligibility damaged for native *and* non-native listeners, roughly equally. Section 1.3 explains the mechanism.

Sentence-level prominence changes recall and social judgement. [E] Hahn (2004) had American undergraduates hear the same lecture with primary stress correctly placed, misplaced, or absent. With correct placement they recalled significantly more content and rated the speaker significantly more favourably. The second half of that matters professionally: prosody moves perceived competence, not just clarity.

Comprehensibility and accentedness are different targets with different predictors. [E] Comprehensibility (how much effort the listener spends) tracks prosody, rate, and fluency. Accentedness (how foreign you sound) tracks segmental accuracy most strongly. So: if you want to be *easy to understand*, work prosody. If you also want to *sound American*, you'll additionally need /r/, /θ/, /h/, /ɪ/–/iː/, dark /l/, and the vowel system. Your priority order already reflects this.

Prosody training may transfer to segments; the reverse is less shown. [T] Some studies report that prosody-focused training improves segmental production as a side effect, without the converse holding. The mechanism is mechanical rather than mysterious: correct rhythm *forces* vowel reduction, linking, flapping, and correct duration, dragging a lot of segmental behaviour with it.

Listeners consciously notice prosody. [T] Asked why someone sounded foreign, listener-judges spontaneously mention rhythm, stress and melody more than specific sounds.

1.3 How prosody carries word boundaries

This is the mechanism section, and it's the most persuasive argument in the document for doing the unglamorous rhythm work.

English speech has no gaps between words. So how does a listener find the word boundaries?

English listeners use strong syllables as boundary hypotheses. [E] Cutler & Norris (1988) proposed the **Metrical Segmentation Strategy**: listeners treat the onset of any *strong* syllable — one with a full, unreduced vowel — as a likely word beginning. The evidence:

- **Word-spotting.** It is harder to detect the word *mint* inside the nonsense word *mintayf* (two strong syllables) than inside *mintef* (strong then weak). When the second syllable is strong,

listeners segment *before* it, so finding *mint* requires reassembling material across a hypothesised boundary. Replicated with American English listeners and materials.
- **Why the strategy pays off.** Cutler & Carter (1987) showed that the large majority of English content words begin with a strong syllable. So "assume a strong syllable starts a word" is a good bet in English. It is a terrible bet in French, where prominence sits at the *end* of the phrase — which is exactly why French listeners develop a different segmentation strategy and why your habits are what they are.
- **Slips of the ear.** Cutler & Butterfield (1992) low-pass filtered the phrase *conduct ascents uphill* — removing most segmental detail but preserving rhythm — and listeners reported hearing *the doctor sends the bill*. Same rhythm, different words entirely. Rhythm alone constrained the parse.

That last example is worth sitting with, because it is the cleanest possible demonstration of what you're doing to your listeners. **When you give an American English with evenly-timed syllables and full vowels everywhere, you aren't giving a slightly-accented signal. You are giving one where every syllable looks like a word onset.** The listener's cheapest parsing heuristic returns garbage and they have to fall back on lexical and syntactic inference. That is the "his English is perfect but he's tiring to listen to" experience, precisely located.

Two corollaries worth stating explicitly:

- **Reduction is not cosmetic; it is the boundary signal.** Every schwa you fail to produce is a false word boundary you insert.
- **Correct word stress is not merely tidy; it's how lookup works.** English listeners key into the lexicon partly via stress pattern. Wrong stress → momentary lookup failure → the listener's processing stalls. This is why Field's result is so strong.

1.4 The four jobs intonation does

Worth keeping separate, because they need different practice.

1. **Grammatical / structural.** Where phrases end, what groups with what, whether it's a question, whether a relative clause is restrictive. `My brother | who lives in Boston | is a doctor` (one brother, plus an aside) vs `My brother who lives in Boston | is a doctor` (identifying which brother).
2. **Informational.** What's new, contrastive, or background. The nucleus system, Part VI.
3. **Attitudinal.** Enthusiasm, doubt, irritation, politeness, warmth. Underdeveloped here is why competent non-native speakers get read as cold or blunt.
4. **Interactional.** Managing the conversation: "I'm not finished", "your turn", "I'm listening", "I want to disagree politely". Invisible until you get it wrong, at which point people interrupt you constantly.

Jobs 1–2 are *clarity*. Jobs 3–4 are *being effective and pleasant to deal with*, and once clarity is solved that's where the professional payoff sits.

1.5 The French → English gap

The structural difference

French organises speech into accentual phrases (APs). An AP is roughly a content word with its function words, 2–7 syllables. In the standard autosegmental-metrical account (Jun & Fougeron 2000, 2002), the AP's underlying tonal shape is **/L Hi L H*/** — an early rise (`Hi`) and a final accent (`H*`). Two structurally different things:

- **`H*`** is a genuine **pitch accent**, anchored to the **last full (non-schwa) syllable** of the AP. It is the primary prominence.
- **`Hi`** is a **phrase accent** anchored to the **left edge** of the AP. Its exact location varies: it can peak on the first, second, or occasionally third syllable, or spread as a plateau over the first two or three. **[E]**

Acoustic comparison confirms `H*` is the stronger of the two: higher F0 peak, longer duration, more reliably anchored to a specific syllable, and a more dynamic rise. **[E]**

Crucially: **French has no lexical stress.** Prominence belongs to the *phrase*, not the word. `le développement` is `le dé-ve-lop-peMENT`; `le développement rapide` moves the prominence to `raPIDE` and *développement* loses it entirely. Syllable durations are relatively even and vowels keep full quality.

English does the reverse. Every polysyllabic word carries a stress pattern stored in the lexicon, fixed regardless of position. `deVELopment` is stressed on `-vel-` in every sentence, every position. And unstressed syllables are not merely quieter: their vowels collapse to `ə/ɪ`, they shorten drastically, and they can be deleted outright. On top of these lexical patterns, the phrase then selects one or two stressed syllables to carry actual pitch movement.

> **French:** prominence at the phrase edge · even syllables · full vowels · no lexical stress.
> **English:** lexically fixed stress · radically unequal syllables · reduced vowels · phrase accents chosen from among the lexical stresses.

Nearly everything else in this document follows from that contrast.

The transfer inventory

Most French C1 speakers have eight to twelve of these. Work through honestly.

#	Transfer	What it sounds like	Where
1	Final-syllable prominence within words	`importANT`, `developMENT`, `photograPHER`	3.9
2	No vowel reduction	`pho-to-graph` with three full vowels; `ˈkɒmfɔːtabl` for `ˈkʌmftɚbl̩`	3.10, 2.3
3	Even syllable duration	listener can't find word boundaries; everything equally important, so nothing is	1.3, 4.1
4	Continuation rise on every chunk	endless-list effect, or perceived uncertainty	7.7
5	Pitch movement at edges rather than on accented syllables	the "French singsong"	7.1, 7.7
6	Focus marked by syntax rather than pitch	`c'est MOI qui l'ai fait` habits → prosodically flat English; your point goes unmarked	6.3
7	No deaccenting of given information	listener can't tell what's new	6.6
8	Narrowed pitch range in English	read as flat, cold, unenthusiastic	7.5
9	Liaison / enchaînement transfer	linking in the wrong places, missing it in the right ones	4.4
10	Epenthetic release after final consonants	`I wantə… stopə…` — extra syllables that wreck the rhythm	4.4
11	No internal rhythm in long words	`ˌuniˈversity`, `ˌcharacteˈristic` flattened	3.3
12	Missing final lengthening / pitch reset	long sentences arrive as one block	5.2
13	One or two tunes used for everything	limited expressive and pragmatic range	7.3

Two root faults

Strip it down and there are two:

Fault A — the temporal fault. Syllables too equal, vowels unreduced. Causes 2, 3, 10, 11; makes 1 audible. Physical and mechanical to fix. Biggest single intelligibility win.

Fault B — the anchoring fault. Prominence goes to the edge of the group rather than to the lexically stressed syllable of the informationally important word. Causes 1, 5, 6, 7. Cognitive to fix: you must decide in real time which word matters, and attach the movement to the right syllable inside it.

Everything else in this document serves one of those two. I won't restate them again.

1.6 Stress "deafness": the honest version

This is widely cited and widely over-simplified — usually presented as a flat verdict that French speakers cannot hear stress. The evidence has two sides, and taken together they give a much narrower and more encouraging diagnosis than the headline finding suggests.

The finding

French listeners are poor at discriminating words that differ only in stress location. [E] Dupoux, Pallier, Sebastián-Gallés & Mehler (1997) found French listeners had great difficulty on nonsense-word pairs differing only in stress, where Spanish listeners (whose language has contrastive stress: `'bebe` "drinks" vs `be'bé` "baby") found it easy. The effect:

- **Is not auditory.** It's a *phonological encoding* problem — French speakers detect the acoustic difference in the moment but can't reliably store stress in short-term memory or use it for lexical access. Performance collapses under memory load and with multiple talkers (Dupoux, Peperkamp & Sebastián-Gallés 2001).
- **Is persistent. [E]** Dupoux et al. (2008) tested French speakers who had learned Spanish as adults, including advanced learners explicitly taught about contrastive stress. They performed no better than French monolinguals. Metalinguistic knowledge did not help.
- **Appears attenuated even in simultaneous French–Spanish bilinguals.**
- **Has a typological explanation. [T]** Peperkamp & Dupoux (2002) and Peperkamp, Vendelin & Dupoux (2010) argue that because French stress is fully predictable, French-acquiring infants learn early that stress carries no lexical information and stop encoding it. Languages with less transparent stress systems retain the representation. Called a "fossil marker" of early development.

The counter-evidence

This half is routinely left out, and it changes the practical picture.

French listeners recognise phrase-level prominence perfectly well. [T] Studies directly testing the perception of the French *final* accent found participants had little difficulty deciding whether a word carried it — a result the authors of that work explicitly present as contradicting a blanket notion of stress deafness in French.

French listeners not only perceive the initial accent, they *expect* it. [T] The optional initial accent `Hi` is readily perceived, and perceived **even when its phonetic correlates are suppressed, or when the F0 rise peaks later along the word** — pointing to a *metrical expectation*, a template the listener imposes rather than merely detects.

There is neurophysiological evidence for a stored representation. [T] A mismatch-negativity study reported evidence for a long-term memory representation of, and phonological preference for, the initial accent — presented by its authors as further evidence against blanket stress deafness.

And the term itself is contested. [C] Work explicitly aimed at disentangling the "phonetic" from the "phonological" components questions whether "deafness" is the right word at all, and note that French listeners perform well in some conditions.

The precise diagnosis

Putting both sides together, the accurate statement is narrower and more useful than "you can't hear stress":

> You are **not** insensitive to prominence. You have a working, well-developed prominence system, including a metrical expectation for early accent. What you lack is (a) the ability to encode prominence **lexically** — to store "this word is stressed *here*" as part of the word — and (b) the habit of anchoring prominence to a **syllable inside a word** rather than to a **phrase edge**.

What follows for practice

1. **You will not reliably acquire English stress patterns by exposure.** Ten years of input hasn't done it. Learn them declaratively: rules (Part III) plus a memorised list for your own vocabulary. Treat stress like irregular verbs.
2. **Don't fully trust your self-monitoring.** When you compare your recording to a model and think "close enough", you may be unable to detect a difference that's glaring to an American. This is why *visual* feedback (9.6) and *external* feedback matter more for you than for a Spanish or Italian speaker.
3. **But your perception is trainable, and there's a specific protocol.** Two converging lines of evidence:
   - **Cue re-weighting is learnable. [T]** Recent work on French listeners' weighting of F0 and duration as stress cues found that participants' *language experience predicted the weights they assigned*, and concluded that prosodic learning consists of learning to assign language-appropriate weights to acoustic cues that aren't themselves language-specific. That is: the machinery is there; the settings are wrong; settings can change.
   - **Perceptual training works, with a known design. [E]** High-variability phonetic training — multiple talkers, varied contexts, trial-by-trial feedback — produces medium-to-large gains in L2 perception (a 79-study meta-analysis reports $g \approx 0.92$ pre/post and $g \approx 0.67$ versus control), with long-term retention and some generalisation to novel items. A separate meta-

analysis finds these perceptual gains transfer to *production*, across proficiency levels. Section 9.2 turns this into a protocol you can actually run.
   - Also relevant: a study using **perceptual fading** — exaggerating the duration cue on stressed syllables, then gradually shrinking the exaggeration — found trained French speakers made significantly fewer errors and responded faster on stress discrimination than untrained controls. Exaggerate-then-fade runs through this whole document for that reason.
4. **Budget accordingly.** At least a third of your practice time on perception for the first several months. Part IX is not background reading for you; it's load-bearing.

1.7 The asset you didn't know you had

Descriptions of French-accented English usually list the `accent d'insistance` — the extra beat French adds to a word's *first* syllable for emphasis, `c'est CAtastrophique` — as an error source. It is also, and more importantly, an asset. Look at what you natively possess:

- A **left-edge accent** (`Hi`) realised primarily as an **F0 rise**, with some **lengthening of the syllable onset**. **[E]**
- Whose location is the **first or second syllable** of the phrase. **[E]**
- Whose function is **rhythmic**: it appears to break up long runs of unaccented syllables — a stress lapse. **[T]**
- Which you **perceive robustly** and even **expect**, with evidence of a stored representation. **[T]**

Now compare what English word stress needs:

- Prominence **anchored to a syllable inside the word**, typically the **first** one, since most English content words are initially stressed (Cutler & Carter 1987). **[E]**
- Realised with **pitch movement plus lengthening**.
- Serving, among other things, **rhythmic alternation** — English inserts secondary stresses to avoid long weak runs, exactly the lapse-avoidance job `Hi` does. **[T]**

These are close enough to be recruited. **[I]** The French system already has a left-anchored, rising, lengthening prominence with a rhythmic function. English word stress is a left-anchored, rising-or-falling, lengthening prominence with a rhythmic function. The difference is that in English it's *lexically specified and obligatory* and it *co-occurs with destruction of the neighbouring syllables* — whereas in French `Hi` is optional, weaker than the phrase-final `H*`, and leaves neighbours intact.

What to do with this.

1. **Reframe the goal.** You are not building a new capacity. You are (i) making an optional accent obligatory, (ii) fixing its position lexically per word instead of letting it float over the first two or three syllables, (iii) making it *stronger* than the phrase-final accent instead of weaker, and (iv) adding vowel reduction on the neighbours. Four modifications to an existing system, not a new system.

2. **Use `accent d'insistance` deliberately as a scaffold.** Take an English word with initial stress. Say it as if you were emphatically insisting on it in French. `c'est CAtastrophique` → `it's 'CATastrophic`. `'MAnagement`. `'DIFficulty`. `'NEcessary`. `'CATegory`. `'TEMporary`. `'PHOtograph`. `'DOcument`. `'ARgument`. `'EXcellent`. `'PREsident`. `'DIFferent`. Every one

of these is a word where the French pull is toward the end and the English target is the first syllable — and your native emphatic-initial accent puts prominence exactly where English wants it.

3. **Then do the second half, which is the hard part.** French insistence gives you a strong first syllable *and leaves the rest intact and phrase-finally accented.* English needs the rest destroyed. So the drill is two-step:

   ```
   step 1 (French scaffold):   CA-TA-STRO-PHIQUE!   → strong initial, all syllables full
   step 2 (English target):    'CAT-ə-strə-fic      → strong initial, everything else collapsed
   ```

   Do step 1, then step 2, alternating, ten times. You are using a native motor pattern to place the accent and then subtracting what English doesn't want.

4. **Fix its position lexically.** `Hi` floats over the first two or three syllables in French. English needs it pinned. Drill contrasts where the pin matters — pairs from the same root whose stress sits on *different* syllables, neither of them predictable from "early-ish": `'PHOtograph` / `pho'TOgrapher`; `ad'MINister` / `ˌadmini'STRAtion`; `e'COnomy` / `ˌeco'NOmic`; `'POlitics` / `po'LItical`; `'ANalyze` / `a'NALysis`. The point of these is to break the "prominence goes early-ish, somewhere in the first two" habit and replace it with "prominence goes on the specific syllable this specific word specifies".

5. **When you do want emphasis in English, don't add a beat — widen the excursion.** English emphasis is a *bigger pitch movement and longer duration on the already-stressed syllable*, often a rise-fall. `That's ab'solutely /\TRUE` — big movement on `-so-`, not an extra hammer on `ab-`.

PART II — THE MACHINERY

2.1 The four acoustic parameters

Fundamental frequency (F0). Vocal fold vibration rate, heard as pitch. Adult male speech typically runs ~75–200 Hz, so set an analysis floor around 60–75 Hz and a ceiling around 250–300 Hz. Primary carrier of intonation *tunes*; one of several stress cues. Only present during voiced sounds, so pitch tracks break up during /s f t/ and during creak.

Duration. Heavily exploited in English: stressed syllables longer, phrase-final syllables longer, unstressed syllables compressed to a degree with no French equivalent.

Intensity. Contributes to stress, but the weakest and least reliable cue; native listeners rely on it less than learners expect. Don't try to fix stress by getting louder — you'll produce loudness without length or reduction, which sounds wrong.

Spectral quality. Chiefly *vowel quality*: stressed syllables have full peripheral vowels; unstressed ones centralise to ə, ɪ, or syllabic consonants. Also voice-quality features. **This is the parameter French uses least and English most.**

The asymmetry that defines your accent. English stress = *longer + fuller vowel + pitch movement + slightly louder*. French phrase-final prominence = *longer + pitch movement*, with no vowel-quality dimension at all, because French has no reduced vowels to contrast against. So when you make an English syllable stressed, you reach for length and pitch on the target — but skip the other half of the job, which is **destroying the neighbours**.

> **English stress is a contrast, not an addition. You create it by suppressing, not by pushing.**

2.2 Two classes of English syllable

Strong (full) syllables have a vowel from the full inventory `/iː ɪ eɪ ɛ æ ɑ ɔ oʊ ʊ uː ʌ aɪ aʊ ɔɪ ɝ/`. They can carry stress. Relatively long.

Weak (reduced) syllables have `ə`, a schwa-ish `ɪ`, word-final `i`, or a syllabic consonant `n̩ l̩ r̩ m̩`. Often less than half the duration of a neighbouring strong syllable. Cannot carry stress.

Calibration: **schwa is the most frequent vowel in spoken English by a wide margin [E]**, and in connected conversational speech the longest-to-shortest syllable ratio within a single sentence can easily exceed 4:1 or 5:1 **[T]**. French has a much narrower range, with the *e muet* as its only near-equivalent to a reduced vowel.

This is the mechanical heart of your accent. `photography` with four full vowels reads as foreign however good your consonants. `fəˈtɑgrəfi` — syllables 1, 3, 4 short, colourless, nearly swallowed; syllable 2 long and clear — reads as near-native even with a French /r/.

Where schwa lives: not predictable from spelling, which is the point. `a` in *about*, `e` in *problem*, `i` in *pencil*, `o` in *common*, `u` in *support*, `ou` in *famous*, `io` in *nation*. Spelling gives the letter; stress gives the sound. Which is why you must know the stress to know the vowels, and why one stress error corrupts a word twice.

2.3 The four segmental prerequisites

Prosody is mostly not in the segmental business, with four exceptions. Reduction *depends* on these four, and without them very little in Parts III and IV is physically possible. You cannot produce a reduced syllable without the equipment to reduce into.

(1) A genuine schwa `/ə/` — not `/ø/` or `/œ/`

French speakers reliably substitute a rounded front-central vowel — the vowel of *deux* or *peur* — for English schwa. It's the nearest French item, and it's audibly wrong: it's rounded, it's front, and it's *too long and too clear*.

English `ə` is: **unrounded, central, brief, and lax.** The tongue is at rest. The lips are neutral, not protruded.

How to find it: say `uh` as in a shrug of indifference, with your lips completely relaxed and slightly parted. Or: begin to say `but` and stop after the vowel, then make it shorter and vaguer still. Or:

hold a relaxed silent posture, then just voice it — schwa is close to the vowel your vocal tract makes when you do nothing.

Test: record about, support, problem, common, famous, nation, again, banana. Listen for lip rounding. If you can see your lips move forward in a mirror when saying schwa, it's wrong.

Why it matters for prosody: if your reduced vowels come out as /ø/, they are (a) too long, (b) too distinct, and (c) perceived as full vowels — which means they read as strong syllables, which means they read as word onsets (1.3). Your reduction fails at the level that matters.

(2) The r-coloured schwa /ɚ/

American English is rhotic, and a huge number of weak syllables end in /ɚ/: bett**er**, wat**er**, aft**er**, oth**er**, memb**er**, custom**er**, manag**er**, comput**er**, importan**t**… and stressed/ɝ/inbird, work, learn, first, service, person`.

This is *not* French /ʁ/. It's an approximant with no uvular friction, made either with the tongue tip curled up (retroflex) or the tongue body bunched — both occur and both work. Zero contact, zero scrape.

How to find it: say uh and, without moving your lips or jaw, pull the whole tongue body backward and slightly upward until the sound darkens. That's it. Then say it as a syllable: -ɚ. Then bett-ɚ.

Why it matters for prosody: if you substitute /ʁ/ you introduce audible friction into a syllable that is supposed to be almost inaudible. A weak syllable with a scraping consonant in it is not weak. Also, /ɚ/ is your handle on the American articulatory setting (8.7) — the tongue posture for /ɚ/ is close to the American resting posture.

(3) Syllabic consonants

In American English, the vowel of a weak syllable frequently disappears entirely and the following consonant becomes the syllable: button = ˈbʌt.n̩, bottle = ˈbɑd.l̩, sudden, written, mountain, little, middle, travel, total, personal = ˈpɝs.n̩.l̩, national, certain, often.

French does not do this, and French speakers insert a vowel: ˈbʌtøn. That single inserted vowel converts a two-syllable word into something with three audible beats.

How to practise: say button but *never open your mouth after the* t. Go straight from the tongue-tip closure into a nasal hum, releasing through the nose. bʌt → n̩. Same for bottle: from the flapped d straight into a lateral l, tongue tip never leaving the ridge.

(4) Unreleased and flapped stops

Two things:

Unreleased finals. English phrase-final /p t k b d g/ are frequently unreleased — you make the closure and simply stop. stop, that, back, bad. French releases them, and often with a small trailing vowel: stopə. That trailing vowel adds a syllable to the end of your phrase, exactly where English is putting final lengthening and a boundary tone. It is one of the most rhythmically destructive habits on the list.

Practise: say `stop` and freeze with your lips closed. Hold for a second. Never open. That's the target.

Flaps. Intervocalic `/t/` before an unstressed vowel becomes a quick voiced tap: `water` → `ˈwɑɾɚ`. Covered in 4.4, but note that it is *a rhythmic phenomenon in segmental clothing* — it happens precisely because the following syllable is weak. If you don't flap, one diagnosis is that you aren't weakening the following syllable enough.

Two more segmental items that touch prosody, briefly:

- **`/h/`.** French lacks it, and French speakers both drop it (`'ave` for *have*) and insert it (`hit is` for *it is*). Insertion is worse for prosody, because it blocks the linking (`it‿is` → `ˈɪɾɪz`) that English rhythm depends on.
- **`/ɪ/` vs `/iː/`.** Many weak syllables use `/ɪ/` (`business`, `village`, `image`, `promise`, `medicine`). Substituting a long `/iː/` makes them strong. Same failure mode as the schwa problem.

Budget: perhaps four to six hours total on these four items, once, early. They are prerequisites, not a parallel project, and once they're in place everything in Parts III and IV becomes possible.

2.4 The prosodic hierarchy

```
UTTERANCE / PARATONE         topic-level unit; pitch reset at its start
 └── INTONATION PHRASE       "thought group"; one nucleus, one boundary tone
      └── (intermediate phrase) smaller unit with a phrase accent
           └── FOOT          one strong syllable + following weak ones
                └── SYLLABLE
                     └── SEGMENT
```

- **Foot:** a stressed syllable plus the weak syllables clinging to it. `(Tell me the)(truth about)(yesterday)` — three feet, three beats, wildly different contents, roughly comparable durations. Feeling the foot is how English rhythm gets into your body.
- **Intonation phrase (IP):** carries one complete tune, exactly one nucleus, ends in a boundary tone with final lengthening. The unit you should think and breathe in. Part V.
- **Paratone:** a spoken paragraph. High onset, gradual descent, low slow ending. Part VIII.

2.5 Pitch accents and edge tones

The single most useful theoretical idea here. Modern descriptions of English intonation (Pierrehumbert 1980; Beckman & Pierrehumbert; the ToBI system) analyse a tune not as a continuous melody but as a **sparse sequence of tonal targets** anchored to two kinds of position:

1. **Pitch accents** — anchored to **stressed syllables** of words the speaker chooses to make prominent. Written with *: `H*`, `L*`, `L+H*` (rise into a high target), `L*+H` (low target then rise), `!H*` (downstepped high — lower than the preceding high).
2. **Edge tones** — anchored to **phrase boundaries**. A **phrase accent** (`H-`, `L-`) covers the stretch from the last pitch accent to the edge; a **boundary tone** (`H%`, `L%`) sits at the very edge.

Everything between targets is interpolation — the voice glides. This is why one tune sounds different across utterances of different lengths without being a different tune. It also gives four ways an intonation phrase can end: `L-L%`, `H-L%`, `L-H%`, `H-H%`.

Why this is the key idea for you. Look again at Fault B. In French the important tonal events are anchored to *phrase edges* — the `H*` lands on the last full syllable of the AP. In English they're anchored to *lexically stressed syllables in the middle of the phrase*, and the edges get only a small tone. Speaking English with French anchoring produces melody of roughly the right shape at the wrong locations. That's exactly why native speakers describe French-accented English as "singing" or "going up at the end of everything": the melody is there, attached to the wrong syllables.

2.6 Notation systems

Arrow notation (British tradition: O'Connor & Arnold, Roach, Wells). Marks the nucleus with its movement: `\ / \/ /\`. Prenuclear accents get `'`. The stretch before the nucleus is the *head*; after it, the *tail*; unaccented syllables before the first accent, the *prehead*. Compact and the best for practice. Used here.

Interlinear "tadpole" diagrams. Two horizontal lines for the top and bottom of your range, big dots for accented syllables, small dots for unaccented, positioned by height. Slow to draw, best for actually *seeing* a contour. Worth doing by hand in your notebook.

ToBI. The research standard for American English. A tone tier (`H*`, `L+H*`, `L-`, `H%`) and a break-index tier (`0` fully joined, `1` normal word boundary, `3` intermediate phrase, `4` full intonation phrase). Learn to read it so you can use the literature.

Levels notation (Pike). Numbers 1–4 for pitch levels: `²I ³didn't ¹say that`. Older American materials.

A caveat. [C] ToBI is a model with live controversies. Some phoneticians argue that analysing the rise-fall as `L+H*` followed by a low tone from elsewhere (an "on-ramp" analysis) generates artefacts, and that an "off-ramp" analysis — a falling accent preceded by a low from elsewhere — is cleaner (Gussenhoven). You don't need to care, but if you read the literature and find people disagreeing about labels, that's what it's about. The *phonetics* isn't in dispute.

PART III — STRESS WITHIN THE WORD

3.1 What word stress is

Every English word of more than one syllable has a **stress pattern**: which syllable is primary, which secondary, which reduced. It's part of the word's identity, stored alongside its spelling and meaning, and it does not move when the word moves in the sentence.

`'photograph → pho'tographer → ˌphoto'graphic → pho'tography`

Same root, four patterns, each fixed. Two consequences:

1. **You have to know it, per word.** Strong tendencies exist (3.4, 3.5) but no complete rule set. English stress is partly predictable from syllable weight and suffix, partly a matter of etymological history, partly arbitrary. Lexical data.

2. **It determines the vowels.** Unstressed syllables reduce, so knowing the stress tells you which vowels are full and which are schwa. Get the stress wrong and *every* vowel in the word is wrong too. That's why one stress error sounds so much worse than one would expect.

3.2 The cue-weighting problem

One finding here should change what you practise, and it is not the one most pronunciation material emphasises.

English listeners weight *segmental* cues to stress most heavily. [T] Cooper, Cutler & Wales (2002), using a word-spotting paradigm, found that although English and Dutch mark stress with a similar bundle of cues — acoustic-prosodic (pitch, duration, intensity) plus segmental (full vowel vs reduced vowel) — **Dutch listeners rely more heavily on pitch, while English listeners rely mostly on the segmental cue**, i.e. on whether the vowel is full or reduced.

Read that again, because it's the strongest single argument in this document for how to spend your time. When an American decides which syllable of your word is stressed, the thing they are most attending to is **whether the other syllables have collapsed**. Not how high your pitch went. Not how loud you were. Whether the neighbours are schwa.

So: the fix for word stress is largely the same as the fix for rhythm, which is largely the same as the fix for reduction. Three problems, one lever.

And on the cue *you* should attend to when listening. It is tempting to conclude that duration must be the handle for a French speaker, since it is the cue least entangled with the phrase-level pitch system. The literature is more equivocal than that:

- Cue weighting is L1-dependent and varies a lot across languages: English and Mandarin listeners weight F0 heavily; Russian listeners weight duration heavily and largely neglect pitch. **[E]**
- Whether **French** listeners compensate for their F0 difficulties by leaning harder on duration and intensity is explicitly flagged in the literature as an **open question**. **[C]**
- What *is* supported: cue weighting is **learnable**, and the weights a listener assigns can be predicted from their language experience — prosodic learning consists largely of learning to assign language-appropriate weights to acoustic cues that aren't themselves language-specific. **[T]**

Practical upshot [I]: train on **vowel quality** as your primary perceptual target, because that's what English listeners use and it's a categorical, easy-to-label distinction (full vs schwa) rather than a continuous judgement. Use duration as your secondary target. Treat pitch as the least reliable cue for identifying *word* stress — it's the cue for *phrase* prominence, which is a different thing (Part VI), and conflating them is a common error.

The four correlates, and your failure mode:

Cue	How much English uses it for word stress	How much you probably use it
Full vs reduced vowel quality	heavily — the primary cue for English listeners	**barely at all**
Longer duration	heavily	somewhat
Pitch movement	only when the word is *accented* in the phrase	at phrase ends only
Greater intensity	modestly	somewhat

Say `deVELopment`. Now check: did syllables 1, 3, 4 get *shorter and vaguer*, or stay full? The American version is roughly `dəˈvɛləpmənt` — schwa, STRESS, schwa, schwa — one syllable twice as long as the other three combined. Four crisp syllables with extra force on the second is the French version of an English stress pattern, and it will be heard as an accent even though you "put the stress in the right place".

Diagnostic. Record `possibility`, `administration`, `responsibility`, `characteristic`, `opportunity`, `communication`. Find native audio. Compare the *duration ratio* of longest to shortest syllable. Natives run roughly 3:1 to 5:1. If yours is 1.5:1, that's Fault A, quantified.

3.3 Primary, secondary, and the internal rhythm of long words

- **Primary** (`ˈ`): the strongest syllable. One per word. Where a pitch accent can land.
- **Secondary** (`ˌ`): unreduced, gets a lesser beat, doesn't carry the main accent. Typically two syllables away from the primary.
- **Unstressed**: reduced vowel, no beat.

`ˌuniˈversity` — `u-` is a real beat with full `/juː/`, `-ni-` reduced, `-ver-` primary, `-si-ty` reduced. Two-beat feel.

A useful generalisation from J.C. Wells: **every English word has a lexical stress (primary or secondary) on one of its first two syllables** — which is why so many long words are effectively double-stressed. **[T]** That's a good default for guessing where the secondary goes, and it dovetails with 1.7: your native `Hi` also lands on syllable 1 or 2.

More: `ˌreprəˈsentative`, `ˌcharactəˈristic`, `ˌadminiˈstration`, `ˌinterˈnational`, `ˌdisorganiˈzation`, `ˈcertifiˌcate`.

Why French speakers flatten these. No equivalent structure, so you either give all syllables equal weight or one big final push. Long Latinate words — the ones you use most professionally — come out unrecognisable. `Administration` with five equal syllables and a final push is genuinely hard to parse in real time.

The fix is rhythmic, not analytic. Learn long words as rhythm patterns using nonsense syllables, then map the word on:

```
administration       ˌad-mi-ni-'stra-tion    =  DA-də-də-DAA-də
opportunity          ˌop-por-'tu-ni-ty       =  DA-də-DAA-də-də
responsibility       re-ˌspon-si-'bi-li-ty   =  də-DA-də-DAA-də-də
characteristic       ˌchar-ac-te-'ris-tic    =  DA-də-də-DAA-də
representative       ˌrep-re-'sen-ta-tive    =  DA-də-DAA-də-də
communication        com-ˌmu-ni-'ca-tion     =  də-DA-də-DAA-də
internationalization in-ˌter-na-tio-na-li-'za-tion = də-DA-də-də-də-də-DAA-də
```

Tap the beats. Say the `da-də` pattern five times. *Then* say the word. This bypasses your unreliable stress perception by routing through motor rhythm — and it's the same trick as the `Hi` scaffold in 1.7, from a different angle.

3.4 Predicting stress: the useful tendencies

Word class. [T] - Two-syllable **nouns and adjectives**: usually first syllable. `'table, 'window, 'happy, 'careful, 'problem, 'process, 'purchase, 'image, 'service, 'office, 'palace, 'courage, 'message, 'village, 'climate, 'private, 'surface, 'promise, 'practice, 'notice, 'purpose`. - Two-syllable **verbs**: usually second. `be'gin, de'cide, a'gree, ex'plain, sup'port, sug'gest, re'lease, col'lect, for'get, main'tain, per'form`. - Real exceptions (`'visit, 'answer, 'follow, 'offer, 'happen, 'listen`), but the tendency is strong enough to guess with.

Noun/verb minimal pairs. A closed, high-frequency set:

Noun / adjective	Verb
`'record`	`re'cord`
`'present`	`pre'sent`
`'object`	`ob'ject`
`'subject`	`sub'ject`
`'project`	`pro'ject`
`'contract`	`con'tract`
`'conduct`	`con'duct`
`'conflict`	`con'flict`
`'contrast`	`con'trast`
`'increase` / `'decrease`	`in'crease` / `de'crease`
`'import` / `'export`	`im'port` / `ex'port`
`'insult`	`in'sult`
`'permit`	`per'mit`
`'progress`	`pro'gress`
`'protest`	`pro'test`
`'refund`	`re'fund`
`'transfer`	`trans'fer`
`'update`	`up'date`
`'upset` (adj)	`up'set`
`'suspect`	`sus'pect`
`'defect`	`de'fect`
`'desert`	`de'sert`
`'combat`	`com'bat`
`'convert`	`con'vert`
`'produce` (= vegetables)	`pro'duce`
`'rebel`	`re'bel`
`'address` (US often)	`ad'dress`

Drill in sentences: *We need to record the record. / They contract to a contract. / The increase will increase costs.*

Syllable weight — the residue of the Latin stress rule. [T] With no stress-determining suffix, English tends to stress the **penultimate** syllable if it's *heavy* (long vowel/diphthong, or closed by a consonant), otherwise the **antepenultimate**. `a'genda` (heavy penult) vs `'cinema` (light penult → antepenult). Predicts a lot of academic vocabulary; good fallback guess.

A productive quirk. [T] Three-syllable words ending in the vowel `[i]` pull strongly toward antepenultimate stress: `'comedy, 'energy, 'strategy, 'remedy, 'quantity`, and the whole `-ity` family. Experimental work (Moore-Cantwell) shows speakers apply this productively to new words, so it's a live pattern, not a historical residue.

Germanic vs Latinate. [T] Short everyday Germanic words are overwhelmingly first-stressed. The long Latinate vocabulary follows suffix rules. Since French shares that Latinate vocabulary, this is exactly where you'll guess wrong from French — see 3.9.

Prefixes are usually unstressed in verbs (`un'do, re'view, dis'cuss, ex'pect, pre'pare`) but often stressed in the corresponding nouns (`'preview, 'overview, 'input, 'output, 'upgrade`), and can take stress for contrast (`I said 'REview, not 'PREview`).

3.5 Suffixes: the three-way taxonomy

The most learnable and highest-yield rule system in English stress.

Class 1 — Stress-neutral (stress doesn't move)

`-ing, -ed, -s/-es, -er, -est, -ly, -ness, -ment, -ful, -less, -able/-ible (mostly), -ish, -hood, -ship, -dom, -ward, -some, -ism, -ist (mostly), -ize (mostly), -en, -age (mostly)`

`'wonder → 'wonderful → 'wonderfully`; `suc'cess → suc'cessful`; `re'ly → re'liable`; `'comfort → 'comfortable`; `'nation → 'nationalism`.

Traps. `'comfortable` = about three syllables (`'kʌmf.tɚ.bəl`), and the French instinct is `comforTABLE`. Likewise `'vegetable` (`'vedʒ.tə.bəl`), `'preferable`, `'comparable`, `'admirable`, `'formidable`, `'hospitable` (US usually `'hɑspɪtəbəl`), `'applicable` (both `'æplɪkəbəl` and `ə'plɪkəbəl` occur in US usage).

Drift. [T] `-able`, `-ory` and `-ly` are moving toward stress-affecting behaviour. American English shifts stress in `ˌneces'sarily, ˌmili'tarily, ˌtempo'rarily, ˌvolu'ntarily, ˌmomen'tarily` — which surprises learners badly, since the base words are all far-left-stressed.

Class 2 — Auto-stressed (the suffix takes the stress)

Many are French loans, so the stress lands where French puts it. These are the easy ones.

Suffix	Examples
`-ee`	`emplo'yee, trai'nee, refu'gee, addres 'see, nomi'nee, guaran'tee` (not `'coffee, com'mittee, 'toffee`)
`-eer`	`engi'neer, volu'nteer, ca'reer, pio 'neer`
`-ese`	`Japa'nese, Portu'guese, journal'ese`
`-esque`	`pictu'resque, grot'esque, Roman'esque`
`-ette`	`kitchen'ette, roul'ette, ciga'rette` (US often `'cigarette`); not `'palette, 'etiquette`
`-ique`	`tech'nique, u'nique, an'tique, cri 'tique, bou'tique`
`-oon`	`bal'loon, car'toon, sa'loon, after'noon`
`-aire`	`million'aire, question'naire, doctri 'naire`
`-self/-selves`	`my'self, them'selves`
`-ade`	`lemo'nade, block'ade, pa'rade, fa'çade` (but US `'decade, 'marmalade`)

Class 3 — Stress-fixing (the suffix determines position in the stem)
Where most of your professional vocabulary lives.

3a — stress on the syllable immediately BEFORE the suffix:

Suffix	Examples
`-ic(s), -ical`	`eco'nomic, ath'letic, scien'tific, stra'tegic, po'litical, his'torical` (exceptions: `'Arabic, 'Catholic, 'politics, 'lunatic`)
`-ity, -ety`	`a'bility, ˌpossi'bility, uni'formity, se'curity, com'plexity, va'riety, au'thority`
`-ion` and its family	`de'cision, ˌadmini'stration, ˌcommuni'cation, 'nation, poli'tician, es'sential, in'dustrial, 'various, ˌadvan'tageous, 'actual, fa'miliar`
`-ify / -efy`	`'clarify, i'dentify, 'justify, 'simplify`
`-itis, -osis, -escence`	`ar'thritis, hyp'nosis, ado'lescence`

The `-ion` rule, stated precisely, is the single highest-yield rule in English stress: an ending consisting of `i` or `u` + one or more vowels + zero or more consonants places the stress on the syllable immediately preceding it, regardless of whether the ending is pronounced as one syllable or two. That covers `-ion, -ian, -ial, -ious, -eous, -ual, -uous, -ium, -ius, -iar, -ience, -iency`. Learn it once, get several thousand words.

3b — stress two syllables before the suffix:

Suffix	Examples
`-ate` (verbs, 3+ syllables)	`'calculate, 'motivate, con'gratulate, 'estimate, ne'gotiate, 'demonstrate, com'municate, par'ticipate`
`-graphy / -grapher`	`pho'tography, bi'ography, ge'ography, pho'tographer`
`-logy / -logist`	`psy'chology, tech'nology, bi'ology, metho'dology, a'pology`
`-cracy`	`de'mocracy, bu'reaucracy`
`-meter`	`ther'mometer, pa'rameter, ki'lometer` (US also `'kilometer`)
`-itude, -itute`	`'attitude, 'institute, 'substitute, 'gratitude`
`-al` on longer stems	`o'riginal, tra'ditional, 'critical`

There are roughly a thousand English `-ate` verbs, pervasive in business and technical English. **Stress two syllables before the `-ate`; pronounce `-ate` as full `/eɪt/` in verbs.** Learners very commonly stress the `-ate` itself. It stays put under `-ed`/`-ing` (`ˈestimated, ˈestimating`) but moves under `-ion` (`ˌestiˈmation`).

3c — the `-ate` noun/adjective split. Same stress position, but `-ate` *reduces* to `/ət/`:

Verb (`/eɪt/`)	Noun / adjective (`/ət/`)
`ˈsepəreɪt` (to separate)	`ˈsepərət` (a separate room)
`ˈestɪmeɪt`	`ˈestɪmət` (an estimate)
`ˈgrædʒueɪt`	`ˈgrædʒuət`
`ˈdeləgeɪt`	`ˈdeləgət`
`əˈproʊprieɪt`	`əˈproʊpriət`
`ˈædvəkeɪt`	`ˈædvəkət`
`ˈmɑdəreɪt`	`ˈmɑdərət`
`əˈsoʊʃieɪt`	`əˈsoʊʃiət`
`dɪˈlɪbəreɪt`	`dɪˈlɪbərət`
`ˈduplɪkeɪt`	`ˈduplɪkət`
`ˈɔltɚneɪt`	`ˈɔltɚnət`
`ˈɪntɪmeɪt`	`ˈɪntɪmət`

An excellent drill, because it makes reduction a *meaningful contrast* rather than an aesthetic preference.

3d — the `-ary/-ory/-ery/-ony` family. These push stress far left, then reduce everything after it, producing very lopsided rhythms French speakers systematically flatten:

`ˈnecessary (ˈnesəseri), ˈsecretary (ˈsekrəteri), ˈtemporary, ˈordinary, ˈcategory, ˈterritory, ˈdictionary, ˈmilitary, ˈinventory, ˈlaboratory (ˈlæbrətɔri — a whole syllable gone), ˈcemetery, ˈstationary, ˈFebruary, ˈmonastery, ˈceremony, ˈtestimony, ˈmatrimony`.

Compare `nécessaire, secrétaire, temporaire, catégorie, territoire, dictionnaire, militaire, laboratoire, cimetière, cérémonie` — every one late-prominent. A concentrated minefield. Drill it as a block.

3.6 Stress-shift families

Learn as families, not isolated words — it's how the system becomes intuitive, and it directly attacks the "prominence floats early-ish" habit from 1.7.

Root	Family
photo-	`ˈphotograph → phoˈtographer → ˌphotoˈgraphic → phoˈtography`
econom-	`eˈconomy → ˌecoˈnomic → ˌecoˈnomical → eˈconomist`
polit-	`ˈpolitics → poˈlitical → ˌpoliˈtician → ˈpolicy`
analy-	`aˈnalysis → ˈanalyze → ˌanaˈlytical → ˈanalyst`
administ-	`adˈminister → adˈministrative (US) → ˌadminiˈstration → adˈministrator`
compet-	`ˈcompetent → ˈcompetence → comˈpetitive → ˌcompeˈtition → comˈpete`
product-	`proˈduce → ˈproduct → proˈduction → ˌproducˈtivity → proˈductive`
technol-	`ˈtechnical → techˈnology → ˌtechnoˈlogical → techˈnician`
origin-	`ˈorigin → oˈriginal → oˌrigiˈnality → oˈriginally`
public-	`ˈpublic → ˌpubliˈcation → ˈpublicize → pubˈlicity`
prefer-	`preˈfer → ˈpreference → ˈpreferable → ˌprefeˈrential`
stable	`ˈstable → staˈbility → ˈstabilize → ˌstabiliˈzation`
person	`ˈperson → ˈpersonal → ˌpersoˈnality → ˌpersonˈnel`
finance	`ˈfinance / fiˈnance → fiˈnancial → ˌfinanˈcier`
industry	`ˈindustry → inˈdustrial → inˈdustrialize → inˌdustriˌaliˈzation`
democ-	`deˈmocracy → ˈdemocrat → ˌdemoˈcratic`
certify	`ˈcertify → cerˈtificate → ˌcertifiˈcation`
examine	`exˈamine → ˌexamiˈnation → exˈaminer`

Method. Say all members in a row, tapping the beat. Then one sentence each. The point is to make the *shift* feel normal.

3.7 Compounds: the honest picture

Textbooks present left-stress as a rule with a short exception list. The literature is blunter than that: Plag's work on the subject is titled *The English compound stress myth.*

What's actually established. [E] Noun–noun compounds in English show **substantial and systematic variation** in stress assignment. `'courtroom`, `'watchmaker` are left-stressed; `silk 'tie`, `Madison 'Avenue`, `singer-'songwriter` are right-stressed. Experimental work with native American English speakers, measuring novel and existing compounds acoustically, finds real variation and support for several competing predictive factors at once — structural, semantic, and analogical. Stress assignment is, to a degree, genuinely unpredictable.

The classic argument that shows the problem. The standard textbook pairs are `Madison 'Avenue` vs `'Madison Street`, and `apple 'pie` vs `'apple cake`. If the second member of each pair is a compound because it's left-stressed, the first must be a compound too — the semantic relation between the two nouns is identical. So left-stress can't be *definitional* of compoundhood.

What predicts left stress. [T] - The compound is **written as one word** (`'keyboard`, `'database`). - It's **frequent** and well established in the language. - It's **semantically opaque** — the meaning isn't derivable from the parts. - In short: **lexicalisation favours left stress**, and combinations may start right-stressed and shift leftward as they become fixed items.

Semantic relations that favour RIGHT stress. [T] This is the useful list — far more reliable than a blanket rule:

Relation	Examples
Material / made-of	`silk 'tie, aluminum 'foil, paper 'bag, apple 'pie` (but `'apple cake`, `'apple sauce`, `'orange juice` — the `-cake` and `-juice` items go left)
Temporal	`summer 'night, morning 'coffee`, `Sunday 'dinner`
Locative / place	`Boston 'Marathon, Madison 'Avenue` (but `'Madison Street` — *street* specifically takes left stress while other road-words take right)
Created-by / authored-by	`a Shakespeare 'sonnet, a Mahler 'symphony`
Copulative (both members equally denote)	`singer-'songwriter, geologist-as'tronomer`

What to actually do about this [I]:

1. **Keep the left-stress default for noun+noun.** It's still the majority pattern and a reasonable guess: `'keyboard, 'airport, 'bedroom, 'database, 'software, 'business plan,`

`'project manager, 'sales report, 'customer service, 'market share, 'price increase, 'cash flow, 'stock market, 'deadline`.

2. **Run the semantic check.** If the relation is material, time, place, authored-by, or copulative, suspect right stress and verify.
3. **Verify rather than derive, for anything you say often.** A dictionary or YouGlish check on your fifty most-used compounds is worth more than any rule.
4. **Don't over-invest.** Getting a compound's stress wrong is far less damaging than getting a *word's* stress wrong, because the listener still has both content words intact. This is a section to be aware of, not to drill for hours.

The genuine minimal pairs are still worth knowing, because here the contrast is semantic:

Compound (left)	Phrase (right)
`'GREENhouse` (for plants)	`green 'HOUSE` (a house that's green)
`'BLACKboard`	`black 'BOARD`
`the 'WHITE House`	`a white 'HOUSE`
`'DARKroom` (photography)	`dark 'ROOM`
`'ENGlish teacher` (teaches English)	`English 'TEACHer` (is English)
`a 'HOTdog` (food)	`a hot 'DOG` (an overheated animal)
`a 'MOVing van` (for moving house)	`a moving 'VAN` (one in motion)

Phrasal verbs are a small, high-frequency, reliably patterned set — worth drilling: verb = right-stressed, noun = left-stressed. `to take 'OFF` / `a 'TAKEoff`; `to break 'DOWN` / `a 'BREAKdown`; `to set 'UP` / `a 'SETup`; `to check 'OUT` / `the 'CHECKout`; `to work 'OUT` / `a 'WORKout`; `to back 'UP` / `a 'BACKup`.

Three-part compounds apply left-stress recursively: `'LIFE insurance salesman`, `'PROject management software`, `'CUStomer service representative`.

Why this section matters to you at all: in French the whole compound gets one final prominence regardless (`assurance-VIE`, `carte de créDIT`), so the left/right distinction has no counterpart and you won't notice you're collapsing it. Professional English is dense with noun compounds.

3.8 Stress shift / iambic reversal

Some words have two stressable syllables and swap which is primary, for rhythmic reasons. **[T]**

`thir'TEEN` alone and phrase-finally (`I'm thir'TEEN`), but `'THIRteen 'MEN` before a stressed word. Same for `four'TEEN … nine'TEEN`, and `ˌJapa'NESE → 'JAPanese 'GARden`, `ˌafter'NOON → 'AFternoon 'TEA`, `ˌaca'DEMic → 'ACademic 'YEAR`, `ˌVietna'MESE`, `ˌIndone'SIAN`.

The principle is **eurhythmy**: English dislikes adjacent heavy beats and dislikes long unbroken runs of weak syllables, and rearranges to alternate. It's why English rhythm has a pulse.

Note also `'THIRty` vs `thir'TEEN` — a real intelligibility trap for numbers. See 3.11.

3.9 The French cognate trap

English and French share tens of thousands of Latinate words. The meanings transferred; the stress didn't. Because French puts prominence at the end and because these words *look* familiar, French speakers systematically produce them with late prominence — and these are precisely the words used in professional English, so the error is maximally visible where you care most.

Mark this up, record yourself reading it, check every one.

Word	English	French pull	Note
important	`im'PORtant`	`impor-TANT`	the canonical error
development	`de'VELopment`	`développe-MENT`	
interesting	`'INtresting`	`intéres-SANT`	often 3 syllables
comfortable	`'COMFtɚbl`	`confor-TABLE`	3 syllables
vegetable	`'VEJtəbl`	`végé-TAL`	3 syllables
restaurant	`'RESTrənt`	`restau-RANT`	2–3 syllables in US
chocolate	`'CHAWklət`	`choco-LAT`	2 syllables
necessary	`'NEsəseri`	`néces-SAIRE`	
secretary	`'SEKrəteri`	`secré-TAIRE`	
temporary	`'TEMpəreri`	`tempo-RAIRE`	
ordinary	`'ORdneri`	`ordi-NAIRE`	
category	`'CATəgori`	`caté-GORIE`	
laboratory	`'LABrətori`	`labora-TOIRE`	US loses a syllable
dictionary	`'DIKshəneri`	`diction-NAIRE`	
military	`'MILəteri`	`mili-TAIRE`	
territory	`'TERətori`	`terri-TOIRE`	
inventory	`'INvəntori`	`inven-TAIRE`	
ceremony	`'SERəmoni`	`cérémo-NIE`	
comparable	`'COMpərəbl`	`compa-RABLE`	
preferable	`'PREFərəbl`	`préfé-RABLE`	
admirable	`'ADmərəbl`	`admi-RABLE`	
formidable	`'FORmɪdəbl`	`formi-DABLE`	
image	`'IMɪj`	`i-MAGE`	2nd syllable `/ɪdʒ/`
courage	`'KURɪj`	`cou-RAGE`	
message	`'MESɪj`	`mes-SAGE`	
village	`'VILɪj`	`vil-LAGE`	
manage	`'MANɪj`	`mana-GER`	
garage	`gə'RAHZH` (US)	`ga-RAGE`	one where US *is* late-stressed
climate	`'KLAImət`	`cli-MAT`	
private	`'PRAIvət`	`pri-VÉ`	
[illegible]	[illegible]	[illegible]	[illegible]

Ones where French serves you well: `per'cent`, `rou'tine`, `ma'chine`, `po'lice`, `ho'tel`, `u'nique`, `tech'nique`, `guaran'tee`, `bal'let`, `ca'fé`, `engi'neer`, `Japa'nese`.

The general heuristic [I]: *if an English word looks French and you don't know its stress, guess earlier than feels natural.* You'll be right far more often than not, because the French pull is always rightward.

How to use the table. Don't memorise it in one go. Read it aloud once with the marks; extract the ~30 in your professional vocabulary and drill those to automaticity; then add words as you catch yourself. A running "my stress errors" list in a note app, reviewed weekly, is worth more than the whole table.

3.10 Vowel reduction: the consequence

Once stress is placed, reduction follows. This deserves separate practice from placement, and per 3.2 it's the cue English listeners weight most.

Five degrees, least to most drastic:

1. **Full vowel** — the stressed syllable.
2. **Reduced to ə/ɪ** — the standard case: `pho'tography` ⟶ `fə'tɑgrəfi`.
3. **Syllabic consonant** — the vowel vanishes: `button = 'bʌt.n̩`, `bottle = 'bɑd.l̩`, `mountain`, `personal` = `'pɝs.n̩.l̩`. (See 2.3.)
4. **Syllable deletion** — a whole syllable goes, in normal speech: `'comfortable` (4⟶3), `'vegetable` (4⟶3), `'chocolate` (3⟶2), `'family` (`'fæmli`), `'every`, `'camera`, `'different` (`'dɪfrənt`), `'business`, `'average`, `'several`, `'general`, `'temperature` (`'tɛmpɚtʃɚ`), `'laboratory`, `'February`, `'probably` (`'prɑbli`), `'interesting`.
5. **Whole-word reduction in function words** — 4.3.

Deletion is not sloppiness. `'comfortable` as four syllables isn't "more careful English"; it's wrong. Americans don't say it that way even formally. Learners over-articulate here because they've been taught clear speech = full pronunciation. In English, clear speech = correct *relative* pronunciation, which includes correct destruction of weak syllables.

Drill reduction independently of placement. Say only the stressed syllable, loud and long. Then add the weak syllables as fast and quietly as physically possible without letting them lengthen. `TAH … TAHgrəf … fə'TAHgrəfi`. Build outward from the beat. This is the reverse of reading a word left-to-right giving each syllable its due.

3.11 Numbers, dates, acronyms, and names

Highly practical, routinely botched, and almost never taught.

Numbers

The teens/tens trap. [E-adjacent, practically critical] `13/30, 14/40, 15/50, 16/60, 17/70, 18/80, 19/90`. Americans distinguish these by **stress position plus the flap**: `thir'TEEN` (final stress, clear `/t/`) vs `'THIRdy` (initial stress, flapped `/t/` ⟶ sounds like `/d/`). If you say `thirty` with a crisp `/t/` and even stress, it will be heard as `thirteen` in half of contexts. If you deal in figures, dates, or quantities professionally, drill this pair set to automaticity — it's the single highest-consequence-per-minute drill in this document.

Long numbers are chunked, and each chunk gets a tune. Phone numbers, account numbers, and codes are grouped in twos and threes, with a **rise on each non-final group and a fall on the last**:

```
415 · 555 · 2 / 1 / 9 / 8      →  four-one-/FIVE | five-five-/FIVE | two-one-
nine-\EIGHT
```

Reciting a long number on one flat tune is a common learner habit and makes it very hard to transcribe. Also: American English says digits individually (`two-one-nine-eight`), not in pairs like French (`vingt-et-un, quatre-vingt-dix-huit`) — and `zero` is often `oh` in strings.

Large numbers. `2,400` = `'twenty-four 'HUNdred` as often as `two 'thousand four 'HUNdred`. Stress the numeral, not the unit: `two 'MILlion 'dollars` has the nucleus on `'DOLlars` or on `'MILlion` for emphasis, but never on `two` unless contrasting.

Decimals and percentages. `3.5%` = `three 'point 'five per'CENT`, with `point` unstressed and quick. `per'cent` has final stress.

Ranges and comparisons take contrastive accent. `We went from 'THIRty to 'FORty` — both numerals accented, because they're contrasted. This is one of the most frequent places nucleus placement matters in professional speech, and one of the easiest to get right once you notice it.

Dates

`'February the 'TENTH` / `February 'TENTH` — nucleus on the ordinal. `'twenty 'twenty-\SIX`. Note `'February` = `'febjueri` in most American speech, with the first `r` commonly dropped. Days of the week: `'MONday` — initial stress and a reduced second syllable, `-di` not `-day`, in fast speech.

Acronyms and initialisms

Letter-by-letter initialisms take stress on the LAST letter: `the U-S-\A`, `the B-B-\C`, `C-\EO`, `A-P-\I`, `H-\R`, `K-P-\I`, `S-Q-\L` (or `'SEquel`), `C-R-\M`, `U-\X`. French speakers often stress the first letter or distribute evenly. Getting this wrong is surprisingly disruptive because initialisms are short and carry a lot of load.

Pronounced acronyms follow normal word-stress rules: `'NASA, 'NATO, 'SAAS` (or `sass`), `'JIRA`, `'SCRUM`.

Spelling aloud. Each letter is a separate accented syllable with a rise-rise-…-fall pattern over the group, and letters chunk in threes: `D-U-/PONT`… For `E/I` and `G/J` confusions, Americans use `E as in echo` — worth having a fixed alphabet ready, since French letter names differ enough to cause real trouble on calls.

Proper names, including yours

Your own name. Americans will re-stress a French name to fit English patterns, and they'll do it whether or not you like it. You have three options, and it's worth choosing deliberately rather than by accident: 1. **Say it in French.** Clear, but expect it to be misheard and mis-repeated, and expect to repeat it often. 2. **Say it with English stress and English segments.** Maximally understood, minimally you. 3. **Say it in French but with English *prosodic* framing** — i.e. keep the French vowels and consonants but place it in the sentence with an English accent pattern and a clear nucleus. This is usually the best compromise **[I]**, and it's what most bilinguals converge on.

Whichever you choose, **prepare one fixed, overlearned version** and always use the same one. Improvising your own name differently each time guarantees confusion.

French place and company names. Americans apply English stress: `Pa'RIS`? No — `'PARis`. `'LYON` (often `lee-'OWN`), `ˌMar'SEILLES`, `'CANNES`, `Bordeaux = bor-'DOH`, `Renault = rə-'NAULT`, `L'Oréal = LOR-ee-'al`, `Hermès = er-'MEZ`. You don't have to adopt these, but **know them**, because you need to recognise your own company's name when an American says it, and you need to decide in advance whether you'll match their version or yours.

Rule of thumb for foreign names in English [T]: English tends to shift stress leftward and reduce the unstressed syllables, which is exactly the general pattern of 3.9 applied to proper nouns.

3.12 Word-stress diagnostic protocol

Do this once thoroughly and you have a personal error map.

1. **Build a list of 100 words you actually use at work.** From your own emails, decks, and documents.
2. **Mark the stress from memory, before checking.** You need to know where your intuitions are wrong, not just what's right.
3. **Check each against US audio** (Merriam-Webster, Cambridge US, Forvo for real-speaker variation, YouGlish for words in real sentences). Mark errors.
4. **Sort errors by category** using 3.4–3.7. You will find patterns — probably `-ary/-ory`, `-able`, `-ate` nouns, and compounds.
5. **Drill the error set in carrier phrases**, not isolation. Isolated words carry their own citation prosody and don't transfer. Use fixed carriers: *The ______ was the problem. / We need to talk about the ______. / I think it's ______.*
6. **Re-test cold in two weeks.** Words right on the second cold test have entered your lexicon. Wrong ones go back in the pile.

Repeat monthly with new lists. Over a year that's a few thousand words with verified stress — a large fraction of the professional vocabulary you'll ever need.

PART IV — RHYTHM

Word stress is the highest-priority *knowledge*. Rhythm is the highest-priority *habit*. Get rhythm right and much else follows mechanically, because English rhythm forces vowel reduction, linking, correct duration, and correct relative prominence.

4.1 “Stress-timed vs syllable-timed”: truth and myth

You’ve heard that English is stress-timed (equal intervals between stresses) and French syllable-timed (equal syllables). **The strong version is false and was disproven decades ago [E]**: inter-stress intervals in English are not isochronous, and French syllables aren’t equal either.

What the terms usefully point at:

- **English compresses unstressed syllables far more aggressively than French. [E]** More unstressed syllables between beats → each gets shorter. `'BIG 'TRUCK` and `'BIGgest of the 'TRUCKS` occupy more similar durations than the syllable count suggests. Not identical, but far closer than in French.
- **English has a categorical strong/weak syllable distinction** (full vowel vs reduced); French doesn’t. **[E]**
- **Quantitative rhythm metrics** — variability indices measuring how much consecutive vowel and consonant durations differ — place English and French near opposite ends of a continuum. **[T]** So the perceptual intuition behind the labels measures something real, even though “isochrony” isn’t it.

The formulation to use: *English syllables are wildly unequal; French syllables are nearly equal. English has beats with rubbish in between; French has an even stream with a bump at the end.*

Metaphor that lands for French speakers: French rhythm is a **string of similar beads**; English rhythm is **Morse code** — long and short in patterned alternation. Musically: French is a stream of even eighth notes; English is syncopated quarters, eighths and sixteenths where only the strong beats land on the pulse.

4.2 The beat system

> **In a neutral English utterance, content words get a beat; function words don’t.**

Content (beats): nouns, main verbs, adjectives, adverbs, question words, demonstrative pronouns, negatives, numbers.

Function (no beat, reduced): articles, prepositions, unstressed auxiliaries and modals, personal pronouns, possessive determiners, conjunctions, infinitival `to`.

```
'TELL me the 'TRUTH about 'YESterday.
   1           2             3
də-DA-də-də-DA-də-də-DA-də-də
```

```
The 'MANager of the de'PARTment has been 'ASKing for a 'MEEting.
         1                2                    3            4
```

Notice how much hides between beats: `of the de'PARTment has been` is six syllables and one beat. All five non-beat syllables get squeezed.

The single most productive physical exercise here: take a sentence, tap the beats at a steady 80–100 bpm (metronome app), and fit everything else in. Can’t fit it → you’re not reducing enough. Sounds choppy → you're inserting pauses instead of compressing. Ten minutes daily changes your rhythm within weeks.

4.3 Weak forms

Every high-frequency English function word has a **strong form** (stressed or in isolation) and a **weak form** used the rest of the time — which is most of the time. Strong forms everywhere is one of the loudest markers of a foreign accent, and very common in French speakers because French has no equivalent process.

Word	Strong	Weak (normal)	Example
a	eɪ	ə	ə ˈkɑr
an	æn	ən	ən ˈæpəl
the	ðiː	ðə / ði before vowel	ðə ˈprɑbləm
and	ænd	ən / n̩	ˈbrɛd n̩ ˈbʌdɚ
but	bʌt	bət	
or	ɔr	ɚ	
than	ðæn	ðən	ˈbɛdɚ ðən ˈðæt
that (conj.)	ðæt	ðət	I ˈthink ðət ɪts ˈfaɪn
as	æz	əz	
of	ʌv	əv / ə	ə ˈlɑd ə ˈpipəl
to	tuː	tə / də after a vowel	I ˈwɑnə ˈgoʊ
for	fɔr	fɚ	fɚ ə ˈmɪnɪt
from	frʌm	frəm	
at	æt	ət	
can	kæn	kən / kn̩	aɪ kn̩ ˈduɪt
could / should / would	full	kəd / ʃəd / wəd	
will	wɪl	l̩ / əl	ˈɪdl̩ ˈwɝk (it'll work)
must	mʌst	məst / məs	
have / has / had	full	(h)əv/v · (h)əz/z · (h)əd/d	ˈaɪv ˈsin ɪt
am / are / was / were	full	əm / ɚ / wəz / wɚ	ˈwɛr ɚ ju
be / been	bi / bɪn	bi / bən	
do / does	du / dʌz	də / dəz	ˈwʌdəjə ˈθɪŋk
he / him / his / her	full	i / ɪm / ɪz / ɚ	ˈtɛl ɚ
them	ðɛm	ðəm / əm	ˈtɛl əm
you / your	ju / jɔr	jə / jɚ	ˈwʌts jɚ ˈneɪm
us	ʌs	əs / s	ˈlɛts ˈgoʊ
some	sʌm	səm	
there (existential)	ðɛr	ðɚ	ðɚz ə ˈprɑbləm
just	dʒʌst	dʒəs	

Strong forms return when the word is contrastively stressed (`I said 'TO him, not 'FROM him`), phrase-final (`Who's it 'FOR? = fɔr`), or cited.

Contractions are obligatory in speech, not informal-only. `I'm, you're, he's, it's, we've, they'll, don't, doesn't, didn't, can't, won't, wouldn't, shouldn't, isn't, aren't, wasn't, hasn't, I'd, that's, there's, what's, let's`. Full forms where a contraction belongs read as emphatic or foreign.

Ultra-frequent multi-word reductions:

Written	Spoken
going to (+verb)	`ˈgʌnə`
want to / wants to	`ˈwɑnə / ˈwʌnts tə`
got to	`ˈgɑdə`
have to / has to	`ˈhæftə / ˈhæstə`
used to	`ˈjustə`
ought to	`ˈɔdə`
kind of / sort of	`ˈkaɪndə / ˈsɔrdə`
a lot of	`əˈlɑdə`
out of	`ˈaʊdə`
let me / give me	`ˈlɛmi / ˈgɪmi`
don't know	`ˈdʌnoʊ`
should/could/would have	`ˈʃʊdə / ˈkʊdə / ˈwʊdə`
what do you / what are you	`ˈwʌdəjə / ˈwʌdɚjə`
did you / don't you	`ˈdɪdʒə / ˈdoʊntʃə`
a couple of	`ə ˈkʌpələ`

A caution. Produce these *because the rhythm produced them*, not as memorised slang. People who consciously insert "gonna" into otherwise stiff, fully-articulated speech sound odd. The reductions are a consequence of correct rhythm; practise the rhythm and they appear.

4.4 Connected speech

English words have no edges in speech. French does this too (liaison, enchaînement) with different rules, so you have the habit and the wrong rule set.

Linking C→V. `an‿apple` → `ə ˈnæpəl`; `pick‿it‿up` → `pɪ ˈkɪ dʌp`; `turn‿it‿off` → `ˈtɝ nɪ ˈdɔf`; `find‿out`.

/r/ is always pronounced. American English is rhotic: `far away` = `ˈfɑr əˈweɪ` with a clear r. Watch for dropping final r before consonants (`car park` → `cah park`), a British/French hybrid.

Intrusive glides. Between vowels English inserts /w/ or /j/: go‿on ⟶ goʊwɑn, do‿it ⟶ duwɪt, I‿am ⟶ aɪjæm, see‿it ⟶ sijɪt. Not a glottal stop, not a hiatus. French speakers commonly insert a small glottal stop here.

Flapping. /t/ (sometimes /d/) between vowels where the *following* vowel is unstressed becomes a quick voiced tap: water ⟶ ˈwɑdɚ, better, city, later, little, whatever, computer, pretty, matter, notice, photo, duty. **Across word boundaries too**: not at all ⟶ ˈnɑd ə ˈdɔl, a lot of ⟶ əˈlɑdə, get it ⟶ ˈgɛdɪt, about it, what if, put on, shut up, sort of. Also after /r/: party ⟶ ˈpɑrdi, dirty, thirty, forty. Arguably the single most identifiable American feature — and a rhythmic one, since it depends on the following syllable being weak.

/nt/ reduction. twenty ⟶ ˈtwɛni, internet ⟶ ˈɪnɚnɛt, interview ⟶ ˈɪnɚvju, international, center ⟶ ˈsɛnɚ, winter ⟶ ˈwɪnɚ (near-homophone of *winner*), wanted ⟶ ˈwɑnəd, plenty, gentle, identify. Very American, very frequent in your vocabulary.

Glottal stops before syllabic n. button ⟶ ˈbʌʔn̩, written, certain, mountain, important ⟶ ɪmˈpɔrʔn̩t, cotton, Latin.

Elision in clusters. next day ⟶ ˈnɛks ˈdeɪ, old man ⟶ ˈoʊl ˈmæn, must be ⟶ ˈməs bi, exactly ⟶ ɪgˈzækli, perfectly, friendship, handbag, grandmother.

Assimilation. ten bikes ⟶ tem baɪks, good boy ⟶ gʊb bɔɪ, did you ⟶ ˈdɪdʒu, don't you ⟶ ˈdoʊntʃu, miss you ⟶ ˈmɪʃu, as you ⟶ ˈæʒu, in case ⟶ ɪŋ keɪs.

Degemination. Identical consonants across a boundary become one long one: black cat, good day, bus stop, some more.

What to avoid: epenthesis. I want-ə to stop-ə — the French release with a trailing vowel. English final stops are often unreleased entirely at phrase end. See 2.3(4). This is one of the most rhythmically destructive habits available.

4.5 Eurhythmy

Two related preferences **[T]**:

1. **English avoids adjacent strong beats.** When two collide, one gives way, or a small pause appears, or pitch differentiates them.
2. **English avoids long runs of weak syllables.** Hence secondary stresses in long words, and hence the tendency toward regular beat counts. (And, notably, the same function French Hi performs — see 1.7.)

Practically: when unsure whether to accent a word, check whether the rhythm alternates. English wants DA-də-DA-də or DA-də-də-DA-də-də, not DA-DA-DA and not DA-də-də-də-də-də-DA.

4.6 Tempo and pausing

Two measures: **speaking rate** (syllables/second including pauses) and **articulation rate** (excluding pauses). Fluent-sounding speech has a moderate articulation rate with well-placed pauses. Learners often do the opposite: fast articulation with random pauses, or uniform slow articulation with none.

Pause typology:

Type	Where	Rough length	Effect
Boundary	at thought-group edges	100–500 ms	structural; helps parsing
Rhetorical	before a key word	200–600 ms	emphasis, anticipation
Breath	at major boundaries	300–800 ms	invisible if at a boundary
Hesitation	mid-phrase, unplanned	variable	damages fluency perception
Filled	`uh`, `um`		fine in small quantities, at boundaries

Rules of thumb [P]: - Pause **at** boundaries, never inside a noun phrase or between a verb and its object. - A pause immediately **before** the most important word is a professional-grade emphasis tool. - Never pause after a preposition, article, or auxiliary. `We talked about … the problem` sounds broken; `We talked about the problem … in some detail` sounds deliberate. - Need thinking time mid-sentence? **Lengthen a function word or use a filler at a boundary**; don't stop dead mid-phrase.

Slowing down is not the answer to clarity. French speakers commonly try to fix intelligibility by slowing uniformly, which *worsens* it: weak syllables get longer, the strong/weak contrast flattens, and you remove exactly the cues the listener needs (1.3). **English clarity comes from increasing contrast, not decreasing speed.** If you must slow down, do it by lengthening the *stressed* syllables and the *pauses*, keeping weak syllables just as short.

4.7 French rhythm error list

Record two minutes of spontaneous speech and check each.

Error	How to detect	Fix
Equal syllable durations	measure syllables/second; suspiciously constant	4.2 tapping
Full vowels in weak syllables	listen for clear `/o e a/` where schwa belongs	3.10, 2.3
`/ø œ/` substituted for schwa	look for lip rounding in a mirror	2.3(1)
Strong forms of function words	`tuː` for `tə`, `ænd` for `ən`	4.3
No contractions	`I am`, `it is`, `do not`	force them in reading aloud
Trailing vowel after final stops	`-ə` after `t p k`	2.3(4)
Beat on the last syllable of long words	`developMENT`	3.3, 1.7
Pauses inside phrases	mark pause locations on a transcript	5.3
No final lengthening	boundaries sound abrupt	5.2
No flapping	crisp `/t/` in `water, better, get it`	4.4
Uniform tempo	no rate variation across a sentence	vary deliberately

4.8 Rhythm drills

R1 — Metronome fit. 90 bpm. Tap the beats, fit everything between. Then 70, then 110. The variation matters more than any single tempo.

R2 — Same beats, different filler. Say these three with *identical* total duration:

```
'DOGS 'EAT 'BONES.
The 'DOGS will 'EAT the 'BONES.
The 'dogs would have 'EATen all of the 'BONES.
```

Three beats each. The third has 12 syllables and should barely exceed the first, which has 3. Fault A in one exercise.

R3 — Limericks and nursery rhymes. English children's verse is built on the foot structure of the language. Limericks are ideal (anapaestic, `də-də-DA`). Memorise three, say them daily at speed. Installs the foot pattern below conscious control.

R4 — Backchaining. Build from the end:

```
…MORning.  →  …in the MORning.  →  …about it in the MORning.
→  …we should TALK about it in the MORning.
→  I THINK we should TALK about it in the MORning.
```

English compresses toward the beats, so building from the end preserves the final rhythm instead of running out of energy.

R5 — Weak-form respelling. Rewrite a paragraph phonetically with all reductions — `We're gonna need ə lodə mor dádə frəm the téam bəfor we cən máke ə decísion` — and read your respelling. Ugly, effective.

R6 — Hum-and-tap. Hum a recorded sentence's rhythm on one pitch, tapping beats. Then hum with the pitch contour. Then say the words. Separating rhythm from melody from words is the core decomposition method.

R7 — Function-word deafness test. Have a partner or TTS read sentences with varying reduction; transcribe them. You'll find you sometimes literally can't hear `and`, `of`, `to`. That gap is why you over-articulate them.

R8 — Filtered rhythm test. Low-pass filter a native sentence below ~400 Hz. You'll hear only rhythm and melody. Try to identify what was said. This is Cutler & Butterfield's experiment (1.3) run on yourself, and it will convince you viscerally how much information rhythm carries.

PART V — THOUGHT GROUPS AND PHRASING

5.1 What a thought group is

Also called intonation phrase, tone unit, tone group, breath group, sense group. A stretch of speech that:

- contains exactly **one nucleus**,
- ends with a **boundary tone**,
- ends with **final lengthening**,
- may be followed by a pause,
- corresponds to one chunk of meaning.

It's the unit in which English is planned, produced and understood. Native speakers don't utter sentences; they utter sequences of thought groups, and listeners process them one at a time, clearing working memory at each boundary. **You are controlling when your listener gets to breathe mentally.**

Typical length: **2–7 words**, or 5–12 syllables. Longer in reading aloud or formal delivery; shorter in animated conversation.

5.2 Boundary cues

1. **Final lengthening** — the last syllable, especially its vowel, gets noticeably longer. The most reliable cue and often the only one when there's no pause. French does this too, at AP edges — you have the mechanism, applied at French-shaped locations.

2. **A boundary tone** — fall (`L%`) for completion, rise (`H%`) for continuation or question, fall-rise for "there's more".
3. **Pitch reset** — pitch drifts down within a group (declination) and jumps back up at a boundary. Large reset = major boundary; small = minor.
4. **Pause** — optional. 50–200 ms at minor boundaries, longer at major. Many boundaries have no pause at all.
5. **Phrase-final creak / glottalisation. [T]** Glottalisation is more frequent at prosodically significant locations — phrase boundaries, utterance boundaries, and pitch accents — and rates are higher at full intonation-phrase boundaries than at intermediate-phrase boundaries, and higher utterance-finally than utterance-medially (Redi & Shattuck-Hufnagel 2001). **But** see 7.6 for the important caveat: individual variation in glottalisation rate is enormous, so treat this as something to *recognise*, not something to install.

Also: **initial strengthening** — the first consonant of a new group is articulated more forcefully. Boundaries are marked on both sides.

5.3 Where to break, and where never to

Natural boundaries: after a long subject; before and after subordinate clauses; around parentheticals and non-restrictive relatives; before coordinating conjunctions joining clauses; between list items; around adverbials; around direct address; before reported speech.

Never break: between a determiner and its noun · between a preposition and its object · between an auxiliary and its verb · between an adjective and its noun · inside a compound noun · between a verb and a short object · in the middle of an infinitive.

Test: if you can't say the chunk in one smooth movement with one main accent, it's too big or cut wrongly.

5.4 Phrasing changes meaning

Phrasing	Meaning
`Let's eat | Grandma ||`	inviting Grandma to eat
`Let's eat Grandma ||`	cannibalism
`My sister | who lives in Lyon | is a lawyer ||`	one sister, who happens to live in Lyon
`My sister who lives in Lyon | is a lawyer ||`	several sisters; the Lyon one
`The students | who studied | passed ||`	all studied, all passed
`The students who studied | passed ||`	only the ones who studied
`He didn't leave | because he was angry ||`	he left, for a different reason
`Do you want tea | or /coffee ||`	open list — maybe something else
`Do you want tea | or \COFfee ||`	closed alternatives — pick one
`Those who sold | quickly | made a profit` vs `Those who sold quickly | made a profit`	different claims about who profited

Be able to produce both members of at least five of these on demand. It's proof you control the boundary as an independent variable.

5.5 Chunk size as rhetoric

- **Short chunks** = emphatic, deliberate. `We tried. | We failed. | We learned. ||`
- **Long chunks** = fluent, casual, low-stakes.
- **Varying length** is what makes speech sound alive. Uniform chunk length in a presentation is hypnotic.

Default for professional English **[P]**: medium chunks (5–8 words) with clear boundaries, shortening to 2–4 words for key points.

5.6 The collision problem

A subtle and important interaction, documented specifically in advanced French learners of English.

In English you must mark **two things with the same acoustic resources**: (a) which syllable of a word carries lexical stress, and (b) where the phrase boundary is. French uses those resources — final lengthening and pitch movement — for boundary marking *only*, having no lexical stress to mark.

So a conflict arises whenever a word's lexical stress is *not* final and the word sits at a phrase edge. Research on advanced French learners using words like `com'puter` and `pro'tection` in varied positions found they cope reasonably in some contexts (isolation; end of a simple statement) but struggle badly in others — **specifically at the boundary of a non-final clause and at the end of a rising question**. There, boundary marking wins and lexical stress vanishes: `com'puter` → `compu'ter`. **[T]**

Drill exactly those contexts:

```
The com'PUter | was the problem ||
Did you check the com'PUter? /
pro'TECtion | is the priority ||
Is it about pro'TECtion? /
Are you the 'MANager? /
Is that the de'VELopment team? /
Is it im'PORtant? /
```

The motor skill is **decoupling pitch height from prominence**: rise to the top of your range on a *final unstressed* syllable while keeping the length and prominence on an earlier one. In French they always coincide; in English they frequently don't. Almost nobody teaches this and it's one of the highest-value single drills available to you.

5.7 Phrasing drills

P1 — Mark-up before speaking. Before reading anything aloud, mark boundaries and nuclei. Then read exaggerating boundaries. Then naturally. Five minutes daily with your own emails or slide notes.

P2 — Boundary-only imitation. Listen to a native paragraph and mark *only* boundaries. Compare with your own reading. Native phrasing is often not where you'd guess.

P3 — One breath per group. Read a paragraph taking a full breath at every major boundary and a small one at every minor. Overdo it.

P4 — Same sentence, N phrasings. One sentence, phrased 1/2/3/4 ways. Notice the meaning shifts.

P5 — Ambiguity pairs. The 5.4 set, both members, without a run-up.

P6 — The collision drill. The 5.6 set. Highest-value phrasing drill for a French speaker.

PART VI — PROMINENCE WITHIN THE SENTENCE (THE NUCLEUS)

Word stress errors make individual words hard to recognise. Nucleus errors make your *point* invisible. Hahn's finding — misplaced primary stress reduces listener recall and lowers speaker evaluations — is about exactly this.

6.1 The nucleus

Within a thought group, one accented syllable is the **nucleus** (also: primary stress, sentence stress, tonic syllable, main accent). It is:

- the **last** accented syllable in the group,
- the one carrying the **nuclear tune**,
- the one the listener treats as the **informational focus**.

Before it may come other accents (the **head**). After it is the **tail**, prosodically dead: no accents, pitch low or interpolating.

```
I 'thought the 'second pre'sentation was the \BEST one.
   head accents                              NUCLEUS   tail
```

Exactly one per thought group. Always exactly one.

6.2 Default placement and its exceptions

The default (broad focus). [T] With nothing contrasted, the nucleus falls on the last content word of the thought group — or, in Wells's better formulation, **on or near the last word of the group *that is important for the meaning***. That qualifier does real work.

```
I'm going to the \OFfice.
She bought a new \CAR.
We need to finish this by \FRIday.
```

Function words in the tail get no accent:

```
I 'don't 'know what to \DO about it.
That's the 'one I was \TALking about.
```

The exceptions are systematic. Wells (2006, ch. 3) catalogues them.

(a) Final time/place adverbials are often skipped when not the point:

```
I 'saw him in the \STREET yesterday.
We \FINished it last night.
```

(Compare a genuine time focus: `We finished it last \NIGHT`, answering "when?")

(b) "Empty" nouns don't take the nucleus — `thing(s)`, `people`, `man`, `woman`, `place`, `stuff`, `one(s)` when semantically light:

```
It's a \DIFficult thing to explain.
There were a lot of \ANgry people.
I want the \RED one.
```

(c) Pronouns and indefinites never take the default nucleus:

```
\GIVE it to me.        I \SAW her.        There's \SOMEthing wrong.
```

(d) Nouns followed by an infinitive keep the nucleus:

```
I've got a \LETter to write.       There's 'nothing \MORE to say.
I have some \CALLS to make.
```

(e) Nouns followed by an adjective or participle keep it:

```
That's the 'only \OPtion available.       He was the 'best \CANdidate applying.
```

(f) In `wh`-questions with a light noun, the nucleus can precede it:

`What \KIND of job?`   `What \SORT of problem?`   `How \MANY people?`

(g) Relative clauses often leave the nucleus on the head noun:

`That's the \MAN I was talking about.`   `Here's the \FILE you asked for.`

(h) Transitive phrasal verbs: object then particle → nucleus on the object:

`Can you 'turn the \LIGHT off?`   `I'll 'pick the \PACKage up.`

(i) Event / intransitive sentences: the nucleus may fall on the subject rather than the final verb. [C — genuinely variable]

`The \PHONE's ringing.`   `My \CAR broke down.`   `The \PRESident died.`

This is a tendency, not a rule. The literature is clear that broad-focus intransitives are **genuinely variable** in English: the nuclear accent can fall on the final verb *or* earlier, and proposed conditioning factors include verb type (the unaccusative/unergative distinction), semantics, and pragmatics — with researchers disagreeing about which factor dominates. Both `An 'actress \BURPED` and `An \ACtress burped` are attested for "What happened?" So: know that subject-nucleus is available and common with "something happened to X" verbs, and don't treat it as obligatory.

(j) Idioms have fixed nuclei:

`\NEVer mind.`   `What's the \MATter?`   `It doesn't \MATter.`

6.3 Narrow focus: putting the nucleus where you mean it

Any accentable syllable can take the nucleus, and the choice changes the sentence. The canonical demonstration:

Placement	Implicature
`\I didn't say he stole it.`	Someone else said it.
`I \DIDn't say he stole it.`	I deny saying it.
`I didn't \SAY he stole it.`	I implied/wrote/hinted it, but didn't *say* it.
`I didn't say \HE stole it.`	I said someone else did.
`I didn't say he \STOLE it.`	I said he borrowed/found it.
`I didn't say he stole \IT.`	He stole something else.

Six placements, six sentences. **In French you'd mostly do this with syntax** — `ce n'est pas moi qui ai dit…`, clefts, dislocations. **English does it with pitch and expects you to.** Two consequences:

1. You produce grammatically correct, prosodically neutral English, and your listener misses which part was the point. In a negotiation or technical disagreement that's expensive.

2. You *hear* neutral English where a native heard a specific claim, so you miss implicatures — your comprehension is affected even at C1.

Drill it by writing the implicature in French. Producing all N placements and translating each implicature makes the mechanism explicit and surfaces the French structure your brain currently reaches for.

6.4 Contrastive accent on function words

Function words can carry the nucleus when they're the point, and then take their **strong** form.

```
I said the book is \ON the table, not \UNder it.
It's not \MY problem, it's \YOURS.
I \DID send it.
\HE may not like it, but \I do.
That's the point I \WAS making.
Is it \FROM him or \TO him?
You \CAN'T do that.  /  You \CAN do that.
```

`I \DID send it` deserves attention: emphatic `do`-support exists in English precisely because English needs somewhere to attach a contrastive accent when the verb itself isn't the contrast. French uses `si` / `bien` / intonation instead. Learn to use `do/did/does` emphatically — it sounds very natural and it's a direct substitute for a French structure you already reach for.

On the `accent d'insistance`: see 1.7 for the full treatment. Short version: use its *placement* as a scaffold for initial word stress, but for sentence-level emphasis in English, widen the excursion on the already-stressed syllable rather than adding a beat to the word's first syllable.

6.5 Prenuclear accents, downstep, declination

Prenuclear accents are optional. [T] In mainstream English varieties, accents before the nucleus are not obligatory; they may mark topic or other information-structural functions, and **they may be inserted for rhythmic reasons, usually near the beginning of a phrase.** (Note that this rhythmic-insertion function is the same job French `Hi` does — 1.7 — which means your instinct to put something early in the phrase is not wrong, it's just under-specified.)

You will see rules of thumb quoted with suspicious precision — "one accent per two or three content words" and similar. I can find no empirical basis for such numbers. The honest guidance: the nucleus is mandatory; prenuclear accents are a choice; accenting *every* content word sounds effortful and emphatic; accenting *none* sounds flat.

Downstep (`!H*`). [E] A pattern in which each successive accent peak is lower than the last, stepping down to the nucleus. Sounds controlled and matter-of-fact.

```
The 'FIRST 'thing we 'need to 'do | is 'check the \NUMbers.
  high    lower  lower  lower              lowest, falls to the floor
```

Worth acquiring deliberately for presentations. Downstep is sometimes described as *the* dominant pattern of American declarative speech; the frequency evidence usually cited for that comes from a study specifically about **contradictions**, so treat the strong version of the claim with caution. That it is common and useful is not in doubt.

The "hat pattern". Pitch rises to a high plateau on the first accent, stays high across the middle, falls on the nucleus. Good for a single coherent piece of information delivered with energy.

Declination. [E] Pitch drifts slowly downward across a group even without accents. So an accent late in a group at the "same" phonological height is physically lower than an early one. Automatic in native speech; learners often fail to produce it, leaving a flat line. Practical fix: start groups *higher* than feels necessary so you have room to descend.

6.6 Deaccenting: the given/new system

Anything the listener already has in mind loses its accent. [E] Not politeness — grammar. In mainstream English varieties, a focused constituent obligatorily contains a nuclear accent, while discourse-given words *following* the focus are unaccented. Failing to do this is one of the most consistent markers of non-native prosody, and notably it's a feature that some contact varieties of English don't use at all.

```
A: Where's the report?
B: I \LEFT it on your desk.                ✓
B: I 'left the re'PORT on your \DESK.      ✗ (report is given)

A: Do you like Boston?
B: I \LOVE Boston.                         ✓ (Boston deaccented, in the tail)
B: I love \BOSton.                         ✗ (sounds like contrast with another
city)

A: Who broke the window?
B: \PEter broke it.                        ✓
B: Peter broke the \WINdow.                ✗

A: I've got a new job.
B: What \KIND of job?                      ✓
B: What kind of \JOB?                      ✗
```

Also applies to synonyms, hypernyms, and inferable material:

```
I bought a car last week. The \DAMN thing already broke down.
He wanted to be a pianist, but he wasn't \GOOD enough at the instrument.
We looked at three apartments. The \THIRD one was best.
```

And to repeated predicates:

```
A: Did you finish the analysis?      B: I 'finished \MOST of it.
```

Diagnostic. Record a two-turn dialogue, reading both parts. Check whether your answers repeat full accents on words that were in the question. One of the most reliably detectable French-speaker errors, and among the easiest to fix, because the rule is clean: **given = no accent.**

6.7 Worked example

```
We should probably move the Tuesday meeting to Thursday.
```

Context	Reading
neutral report	`We should 'probably 'move the 'Tuesday meeting to \THURSday.`
answering “which meeting?”	`We should move the \TUESday meeting to Thursday.`
answering “should we cancel it?”	`We should probably \MOVE it, not cancel it.`
hedging	`We should \/PROBably move it.`
contradicting “we can’t move it”	`We \CAN move it.`
proposing tentatively	`We should probably move it to /THURSday?`
exasperated	`We should 'JUST \MOVE the 'damn thing.`

Practise generating these *from a context prompt*, not reading them off. The skill is real-time nucleus selection, which is a comprehension-of-context skill as much as a motor one.

6.8 Prominence drills

N1 — Six-way sentence. All placements of one sentence, with the implicature stated in French. Then invent your own from work vocabulary: `We can't ship the update on Friday.`

N2 — Question→answer nucleus. Write 20 questions and answers; mark the nucleus; check it’s on new information and that given information is deaccented; then say them.

N3 — Contrast pairs. `not X, Y` structures:

```
I said in'CREASE the budget, not de'CREASE it.
It's on the 'THIRD floor, not the 'FIRST.
We need 'MORE data, not 'BETter data.
I'm not 'complaining, I'm ex'PLAINing.
```

N4 — Deaccenting dialogues. Take a real two-person transcript. Mark every word that’s *given* by the time it’s uttered. Read it deaccenting all of them. Will feel unnaturally flat at first; that flatness is what English wants.

N5 — Numbers in contrast. `We went from 'THIRty to 'FORty · It's 'THIRteen, not 'THIRty · Q\THREE, not Q\TWO`. Combines 3.11 with contrastive accent — the single most common professional use of this skill.

N6 — The “so what” test. After every sentence you say, silently ask: which word was the point? Did the nucleus go there? Do this deliberately in real conversation for a week. It’s the bridge from drill to spontaneous speech.

N7 — Nucleus-only shadowing. Listen to a native paragraph. Don’t repeat words — say `DA` on each nucleus at the right pitch and time, silent elsewhere. Isolates the nucleus stream. Then add words back.

PART VII — MELODY: THE TUNE INVENTORY

7.1 Anatomy of a tune

British-tradition anatomy (best for practice, because it tells you what to *do*):

```
(prehead)   HEAD              NUCLEUS      (tail)
 I was      'thinking we      \MIGHT       be able to do it
```

- **Prehead:** unaccented syllables before the first accent. Usually low and fast.
- **Head:** from the first accent up to the nucleus. Contains any prenuclear accents.
- **Nucleus:** the accented syllable carrying the main pitch movement. Mandatory.
- **Tail:** everything after. Continues the nuclear movement; no accents of its own.

ToBI anatomy (best for precision and for reading research):

```
pitch accents on chosen stressed syllables  +  phrase accent  +  boundary tone
   H*      H*        L+H*                          L-                L%
```

The key insight, restated because everything depends on it: the pitch movement happens on the nucleus, not at the end of the phrase. When the nucleus *is* the last syllable, they coincide and you can't tell. When there's a tail, they don't:

```
\NO.                           → the fall happens on "no"
\NO, I don't think so.         → the fall happens on "NO"; everything after is
already low
It's a'BOUT the \BUDget.       → fall on "BUD-"; "-get" already low
It's a'BOUT the \MANagement.  → fall on "MAN-"; three low syllables after
```

Drill this specifically. Say `MANagement` at the end of a statement: the pitch must drop on `MAN-` and stay at the bottom for three syllables. A French speaker's instinct is to hold the voice up until the last syllable and then drop.

```
It's about the \MANagement.
I talked to the \MANager.
That's the \DIFficulty.
It was \COMPletely unnecessary.
We need a \BEDdɚ solution to the problem.
```

Five long-tailed nuclei, each with the fall on the stressed syllable and a flat low tail. This one drill removes a large share of the “French sing-song” impression.

7.2 The compositional system

Most treatments give you a list of ten tunes with a gloss attached to each. That is a weak way to learn them, because you cannot extrapolate from a list — every new contour you meet is a new item to memorise. There is a better framework.

Pierrehumbert & Hirschberg (1990) proposed that **tune meanings are compositional**: they're built out of the meanings of the individual tones. Their claim, in their own framing, is that a speaker chooses a tune to convey a particular relationship between the utterance, the hearer's currently perceived beliefs, and anticipated later contributions — and that these relationships are **composed from the pitch accents, phrase accents, and boundary tones making up the tune.**

The widely-cited element meanings **[T]**:

Pitch accents — what the speaker is doing with the accented item

Tone	Meaning
`H*`	This item is **new** — add it to what we mutually believe. Plain assertion.
`L*`	This item is **salient but not asserted** — mentioned, pointed at, but explicitly *not* being predicated or added. (Hence its role in questions and in the contradiction contour.)
`L+H*`	This item is **contrastive** / evokes a **scale** — it stands against a salient alternative, or corrects one. The excursion is doing the contrast work: American English `L+H*` shows a reliably higher F0 peak than plain `H*`. **[E]**
`L*+H`	Like `L+H*` but the speaker **withholds commitment** — uncertainty, incredulity, scepticism about the scale.
`H+L*` / `H*+L`	The item should have been **inferable** from what we already mutually believe. Often reads as "obviously", sometimes as condescending.

Phrase accent — how this phrase relates to the *next* one

Tone	Meaning
`H-`	**Forward-looking.** Interpret this phrase with respect to a following phrase.
`L-`	No such forward reference. This phrase stands on its own.

Boundary tone — who carries the interpretive burden

Tone	Meaning
`H%`	Direct the hearer's attention **forward**: the utterance is to be completed by the hearer or by subsequent discourse. Passes the burden across.
`L%`	No such direction. The phrase is self-contained.

Why this is worth the effort
Now you can **generate** rather than memorise. Work an example:

- `L*` (salient, not asserted) + `H-` (forward-looking) + `H%` (your turn) = "I'm putting this in front of you, not asserting it, and it's for you to resolve." ⟶ **a yes/no question.** You didn't have to memorise that; it falls out.
- `L+H*` (contrastive) + `L-` (self-contained) + `H%` (over to you) = "here's a contrast, this phrase stands alone, but you complete the inference." ⟶ **the fall-rise**, which is exactly why it means "…but there's something I'm leaving for you to work out."
- `H*` (new, asserted) + `L-` + `L%` (self-contained, burden mine) = "I assert this, complete." ⟶ **the plain falling statement.**
- `H*` + `H-H%` = "I assert this, *and* it's forward-looking, *and* attend to what follows." ⟶ **uptalk**, which is why its main documented function is continuation and turn-holding rather than uncertainty (7.3, tune 3).

The honest caveat. [C] This framework is influential but its semantics has been criticised as informal and not fully supported by the data. One critique notes that although the broad outlines — compositionality of tone meaning, intonation as signalling relations among discourse participants' mutual beliefs — are appealing, the exact semantics remains loose, and boundary tones in particular don't behave uniformly with respect to question identification.

So: **use it as a generative heuristic, not a law. [I]** It will let you predict most tune meanings from their parts and give you a principled reason for each of the tunes below, which is far better than memorising ten glosses. But don't be surprised when a particular contour in real speech doesn't fit cleanly.

A second caveat on how many tunes are real. [C] Studies using imitation and discrimination tasks to test whether the twelve theoretically-predicted nuclear tunes are genuinely distinct categories find that the most robust distinction speakers make is between **rise-only** and **rise-then-fall** trajectories, with finer distinctions less categorically separated. Practical translation: **get the rise/fall distinction rock solid, master five or six tunes, and let the rest come.**

7.3 The core tunes

Tune 1 — The Fall (\) · `H* L-L%`
Shape: up to a high point on the nucleus, then fall to the bottom of your range. **Composition:** asserted + self-contained + burden mine. **Meaning:** completeness, assertion, finality, confidence. Default for statements, commands, wh-questions.

```
I'll 'send it on \MONday.        The 'meeting's at \THREE.
\CLOSE the door.                  'Where are you \GOing?
'What's the \PROBlem?             \THANKS.
```

The commonest learner failure: not going low enough. A fall stopping mid-range reads as unfinished or evasive. Americans commit to the floor. In Praat, check that your final F0 approaches your measured minimum.

High vs low fall. A high fall (starting very high) is emphatic and engaged; a low fall (starting mid) is calm or flat. Both are "the fall". Using only low falls is a major cause of sounding uninterested.

Tune 2 — The Low Rise (/ from low) · `L* H-H%`
Composition: salient-but-not-asserted + forward-looking + your turn. **Meaning:** genuine question, checking, politeness, encouragement.

```
Are you /READy?        Did you /SEND it?        Is that /OK?
/REAlly?               Do you have a /MInute?
```

Yes/no questions default to a rise, as in French — easy for you. But they can also fall, with a shift in meaning: `Are you \READY?` is a command or an impatient prompt; `Are you /READY?` is a genuine inquiry.

Wh-questions default to a FALL, not a rise. `Where are you \GOing?` This differs from French, where wh-questions often rise. A rising wh-question in English means something specific: an echo ("repeat that") or a softened, gentle inquiry.

```
'What's your \NAME?     neutral
'What's your /NAME?     "sorry, could you repeat that?" or very gentle
```

Rising every wh-question makes you sound perpetually unsure or as if you didn't hear the answer. Fix this early; it's cheap and highly noticeable.

Tune 3 — The High Rise / "Uptalk" (/ from high) · `H* H-H%`
Composition: asserted + forward-looking + attend to what follows. Note that the composition predicts *continuation*, not uncertainty — which matches the data.

Documented functions in American English [T]: - **Turn-holding / continuation** — "I'm not finished". Studies find high rises far more common *mid-turn* than turn-finally, and especially frequent in narratives. - **Engagement- and comprehension-checking** — "are you with me?" - **Marking new information** or introducing a topic. - **Stance softening.**

```
So I 'called the /VENdor? | and they 'said the 'part was /BACKordered? |
and 'now we're 'looking at \MARCH ||
```

Two high rises holding the floor, then a fall to close.

Should you use it? [I] Selectively. It's very characteristic of contemporary American speech, so avoiding it entirely reads as slightly formal or foreign. But it's socially marked: perception studies find HRT-inflected statements rated as **less assertive**, and there's evidence of a cost in competitive or hierarchical exchanges. My recommendation: **learn to produce it, use it for continuation and narrative, avoid it on your conclusions and key assertions.** Conclusions fall.

Terminology note. [C] "High Rising Terminal" is a poor label — Mark Liberman has pointed out that many instances of uptalk *start low* in the speaker's range and rise from there, so "high" is often phonetically false. Related work finds the range and alignment of the rise much more variable than early descriptions claimed. Listen for the shape, not the label.

Tune 4 — The Fall-Rise (\/) · `L+H* L-H%`
Composition: contrastive/scalar + self-contained + burden passed to you. Hence: "here's a contrast, and I'm leaving you to work out the rest." **Meaning:** reservation, incompleteness, implication, "but…", politeness.

```
I \/LIKE it…                      (…but there's a problem)
It's \/POSSible.                   (…but unlikely)
\/SOME of them agreed.             (…not all)
We could \/TRY that.               (…though I'm not sure)
I \/THINK so.                      (hedged)
\/YES…                             (reluctant agreement)
\/WELL…                            (polite disagreement incoming)
I didn't say he \/STOLE it.        (I said something else)
That's 'not exactly \/TRUE.
As 'far as \/I know…
```

Also: contrastive topic. `As for the \/BUDget, | we're \FINE.` "Regarding X as opposed to other things."

Why it matters most professionally. This tune is how English speakers disagree politely, hedge, express doubt, signal a caveat, and soften bad news — without saying so lexically. Without it you have two options: bald assertion (fall) or question (rise). Bald assertion where a fall-rise was expected is precisely why non-native professionals get read as blunt or tactless. It is your politeness engine.

How to produce it. Start mid, jump up on the stressed syllable, drop, then float back up. Trace the shape with your hand. It's a bigger, more athletic movement than French speakers habitually make and it will feel theatrical. Overdo it; it normalises.

Tune 5 — The Rise-Fall (/\) · `L+H* L-L%` with a wide excursion
Composition: contrastive + self-contained + burden mine. A contrast, fully asserted. **Meaning:** emphasis, strong assertion, surprise, being impressed. The contrastive-focus tune of American English.

```
That's /\FANtastic!      It was /\HUGE.        I /\TOLD you.
That's /\EXactly what I said.        He's /\BRILliant.
I did /\NOT say that.
```

Emphasis in English = wider pitch excursion + longer duration on the nucleus. Not more volume, not an extra beat on the word's first syllable. The direct replacement for `accent d'insistance` at the sentence level (see 1.7 and 6.4): - `C'est ABsolument inaccepTABLE` ⟶ `That is ab'solutely un/\ACCEPtable.` - `C'est VRAIment important` ⟶ `That's /\REALly important.`

Tune 6 — The Level / Plateau (→) · `H* H-L%` or a flat mid-range
Meaning: routine, non-committal, bored, or "one item in a series". The list tune and the tune of formulaic transactional speech.

```
→NEXT. →NEXT. →NEXT.        Name? →Address? →Date of birth?
Yeah. →Yeah. →Sure.          (perfunctory)
```

Use with caution — flat is already your failure mode. But you need to *produce* it deliberately, and to *recognise* it, because a native giving you flat plateaus is signalling disengagement and you want to notice.

Tunes to recognise now, produce later
The Downstepped Fall (`!H*` series). Each accent lower than the last, ending in a fall. Calm, structured, professional. See 6.5.

The Contradiction Contour (`L* … L*+H L-H%`). Distinctively American. Low and flat through the utterance, then a rise. Used to contradict a presupposition. Composition explains it: `L*` = "this is salient but I'm not asserting it", i.e. "you brought this up, not me", plus a final rise handing the problem back.

```
I'm 'not going to /JaPAN.        (low flat lead-in, final rise: "you've got that
wrong")
I 'don't have a /CAR.
```

Studies of American contradictions find the classic pattern is one or more `L*` accents with a rising nuclear tune, though they also find contradictions can be marked with falls for various reasons — the rise is optional, not obligatory. **[T]** Very useful, and it sounds unmistakably American.

The Calling / Chanted Contour (`H* !H- L%`). A step down to a sustained level. Vocatives, calling across distance, announcing.

```
'MAR-cus!     'DIN-ner!
```

The Surprise-Redundancy Contour. A high rise-fall early, then a step to a level tail: "obviously, I'd have thought you knew." The `H+L*` accent (inferable-from-mutual-belief) is doing the work, which is why it can read as condescending. Recognise it; don't use it.

The Low Flat / “scathing” contour (`L* L-L%`). Everything at the bottom. Bored, grim, dismissive, or dangerous.

```
→→→I ˈdon't ˈcare.        →→ˈThat's ˈgreat.        (sarcastic)
```

7.4 Tunes by sentence type

<table>
<thead>
<tr><th>Type</th><th>Default American tune</th><th>Notes</th></tr>
</thead>
<tbody>
<tr><td>Statement, complete</td><td>\ fall to the floor</td><td>high fall = engaged; low fall = flat</td></tr>
<tr><td>Statement, more coming</td><td>\/ or /</td><td>fall-rise if there’s a caveat; high rise if just continuing</td></tr>
<tr><td>Wh-question</td><td>\ fall</td><td>rise = echo or softened. Key difference from French.</td></tr>
<tr><td>Yes/no question</td><td>/ low rise</td><td>fall = impatient or rhetorical</td></tr>
<tr><td>Echo / repeat request</td><td>/ high rise on the queried element</td><td>You did \WHAT? = astonishment; You did /what? = didn’t hear</td></tr>
<tr><td>Tag question, genuine</td><td>/ rise</td><td>It's cold, 'isn't /it?</td></tr>
<tr><td>Tag question, rhetorical</td><td>\ fall</td><td>It's cold, 'isn't \it.</td></tr>
<tr><td>Alternative question, closed</td><td>rise then fall</td><td>Tea \| or \COFfee?</td></tr>
<tr><td>Alternative question, open</td><td>rise throughout</td><td>Tea \| or /COFfee?</td></tr>
<tr><td>List, closed</td><td>rise on each, fall on last</td><td>'Red \| 'white \| and \BLUE.</td></tr>
<tr><td>List, open</td><td>rise throughout</td><td>'Red \| 'white \| /BLUE…</td></tr>
<tr><td>Command</td><td>\ fall</td><td>\STOP.</td></tr>
<tr><td>Polite request</td><td>\/ or /</td><td>Could you \/CLOSE the door?</td></tr>
<tr><td>Exclamation</td><td>/\ wide</td><td>How /\NICE!</td></tr>
<tr><td>Calling</td><td>calling contour</td><td></td></tr>
<tr><td>Vocative before a statement</td><td>\/ or level</td><td>\/MARC, \| can you check \THIS?</td></tr>
<tr><td>Greeting</td><td>wide, warm</td><td>\/HI! — flat →Hello reads cold</td></tr>
<tr><td>Thanking</td><td>wide \</td><td>flat thanks reads perfunctory</td></tr>
<tr><td>Apologising</td><td>\/ or wide \</td><td>I'm \/SO sorry.</td></tr>
<tr><td>Enthusiastic agreement</td><td>/\</td><td>Ab/\SOLUTEly. Ex/\ACTly.</td></tr>
<tr><td>Reluctant agreement</td><td>\/</td><td>\/Yeah…</td></tr>
<tr><td>Parenthetical</td><td>compressed range, faster, lower</td><td>My brother – 'who lives in Boston – is a \DOCtor.</td></tr>
<tr><td>Number strings</td><td>rise on each group, fall on the last</td><td>see 3.11</td></tr>
</tbody>
</table>

7.5 Pitch range, declination, reset

The standard advice here is that English uses a wider pitch range than French, and that you should therefore widen yours. That framing is wrong in an interesting way, and the correct version points at a different fix.

What the research actually shows:

(1) Cross-language pitch-range comparison is methodologically hard and results are inconsistent. [C] Studies comparing pitch range across languages get different answers depending on the measure used. One systematic comparison of Southern British English and Northern Standard German found **no** span difference on some linguistically-based measures despite the strong stereotype that English is wider, while a difference in the predicted direction appeared on another measure — with the authors concluding that the standard measures may not be suitable for cross-language comparison at all because tonal structures aren't equally distributed across languages. Other work on the same pair *did* find English wider. So: don't trust confident claims either way, including my earlier one.

(2) Specifically on French: French and German speakers showed no pitch-range difference in their *native* production for short read sentences. [T] There isn't good evidence that French is a narrow-range language.

(3) But L2 speech is reliably narrowed. [T] Zimmerer and colleagues (2014), studying French *and* German L2 users of English, found **a consistent narrowing of pitch range in L2 speech**, attributed to reduced confidence and to cognitive load — participants appeared to prioritise lexical and segmental accuracy, which limited their ability to deploy native-like pitch patterns. The same narrowing has been found for other L1s (e.g. Japanese speakers using narrower ranges in L2 English than in L1, for both sexes).

(4) And wider range is rewarded. [T] Work on L2 assessment finds wider pitch range correlating with better ratings and more effective delivery.

So the correct diagnosis, and it is better news than the standard framing:

> Your French pitch range is probably normal. You are **suppressing** it in English, most likely because your attention is consumed by words, grammar and sounds. This is a *load* problem, not a *capacity* problem.

What follows:

1. **Measure both.** This is why the day-one baseline (0.4) asks for your pitch span in *French* as well as English. If your French span is much wider, you have your answer and your target: your own French range.
2. **The fix is partly to reduce load, not to practise pitch.** Every chunk you overlearn (10.2), every word whose stress you've automated (3.12), every phrase you no longer have to plan — each frees attention that can go to pitch. A large part of "widening your range" is achieved indirectly, by making everything else cheaper.

3. **Do also practise it directly**, but understand you're removing an inhibition rather than building a capacity. That's psychologically different and usually faster.
4. **Check *where* the movement is, not just how much.** Even with an identical total span, your movement may sit at phrase edges (French anchoring) rather than on accented syllables (English anchoring). Look at the pitch track and ask: *do the peaks sit on stressed syllables in the middle of phrases?* This is Fault B and it is not fixed by widening.

Declination. [E] Pitch drifts down across a group automatically in native speech. If your pitch is flat across a group and then drops at the end, that's French phrasing. Start groups higher so you have room to descend.

Pitch reset. [E] Pitch jumps back up at the start of each new group; the size of the jump signals the size of the boundary. Small = minor boundary; large = new sentence; very large = new topic (8.1). A genuinely learnable, high-payoff skill for presentations: deliberately reset high at each new point.

Register shifts. Compressed, lowered register = parenthesis, aside, or lower importance. Raised register = importance, excitement, new topic. Native speakers modulate constantly.

7.6 American habits

Features that read as specifically *American* rather than just *English*:

1. **Falls that reach the floor.** See tune 1.
2. **Wide `L+H*` excursions for contrast.** Americans mark contrast with pitch jumps that can feel excessive to a French ear. **[E]**
3. **Downstepping declaratives. [E]**
4. **Frequent high rises mid-turn. [T]**
5. **Heavy flapping and `/nt/` reduction.** Segmental in form, rhythmic in cause.
6. **Rhoticity everywhere**, changing syllable shapes and giving American speech its resonance.
7. **A more retracted, more open, somewhat nasal articulatory setting.** See 8.7 — with caveats.
8. **Fillers carry pitch.** American `um`/`uh` sit at boundaries with a mid-level or slightly falling pitch. A French `euh` transplanted into English is instantly recognisable.

On creak

Accent-training material commonly tells learners to acquire utterance-final creak as an American feature. That is too strong. Here is the accurate picture:

What holds up: - **Phrase-final creak is a real boundary phenomenon in American English. [E]** Glottalisation clusters at prosodically significant locations — phrase boundaries, utterance boundaries, and pitch accents. Rates are higher at full intonation-phrase boundaries than at intermediate-phrase boundaries, and higher utterance-finally than utterance-medially (Redi & Shattuck-Hufnagel 2001; Dilley et al.; Pierrehumbert 1995). - **It's common across varieties of English** — and also attested in other languages, including Spanish.

What doesn't support "you must learn to creak": - **Individual variation is enormous. [E]** Redi & Shattuck-Hufnagel found rates of glottalisation on word-initial vowels ranging from **13%**

to 44% across five professional announcers, and from **under 1% to 34%** across nine speakers reading the *same* text. There is no single "American amount" of creak. - **Gender findings conflict. [C]** Some studies find higher rates in women (one found 38–44% for female radio announcers vs 13–24% for male), others find men creakier by acoustic measures. Reviews describe the results as conflicting, with anatomical, sociolinguistic and structural explanations all in play. - **And crucially for you: a study of 49 Canadian English–French bilinguals found no strong or consistent evidence for cross-linguistic differences in creak**, concluding that variation was better explained by vocal ageing than by language or ongoing sound change. **[T]** If bilinguals carry the same voice quality across both languages, creak is a poor candidate for "an English thing to acquire."

Advice [I]: don't practise creak. Practise **letting your pitch actually reach the bottom of your range at utterance end** (tune 1), which is a genuine and important target. Creak will often appear as a *by-product* of getting there, because that's partly what creak is — phonation becoming irregular as F0 bottoms out. If it appears, fine. If it doesn't, you've still hit the target. What you should absolutely do is **recognise** it, because it's a boundary and turn-yielding cue you need to read in others (8.2).

7.7 French transfer patterns to unlearn

(1) The continuation rise on every group. French marks non-final groups with a rise (*continuation majeure/mineure*). Transferred wholesale:

```
Yesterday I went to the /OFfice | and I 'met the /CLIent |
and we dis'cussed the /CONtract | and 'now we're /WAITing ||
```

Every group identical. Reads as an unending list, nervousness, or a series of questions. **English varies its non-final tunes**: fall-rise, high rise, low rise, mid-level, and sometimes a full fall mid-utterance when the chunks are informationally independent. Fix: consciously vary. Replace every second rise with a fall-rise, and some with a plain fall.

(2) Melody anchored to edges rather than accents. The `MANagement` drill (7.1).

(3) The AP's initial rise landing as a spurious accent. French `Hi` can put a small rise on the first syllable of an English phrase where English wants nothing. Note the asymmetry with 1.7: `Hi`'s *placement* is an asset for word-initial stress; its *automatic, floating, phrase-level* deployment is the liability. The difference is whether it's pinned to a lexically specified syllable or drifting over the phrase onset.

(4) Narrow excursions on accents, big movement at edges. The inverse of English. Fix: exaggerate the jump on nuclei and *reduce* your final movement — except at genuine finality, where you fall hard.

(5) Rising wh-questions. Fall.

(6) Uniform tune choice. Many French speakers settle on one or two tunes for everything. Fix: the 7.3 inventory, one tune at a time.

(7) Flat range as over-correction. Some advanced speakers, told they sound sing-song, suppress prosody and become monotone — trading one problem for a worse one. Don't reduce your movement; **relocate** it.

7.8 Politeness and being misread

Real professional consequences, and nobody warns people.

What you produce	How it's read	What to produce instead
Bald fall on a disagreement: `That won't \WORK.`	blunt, dismissive, arrogant	`That \/might not work. / I'm 'not 'sure 'that'll \/ WORK.`
Flat statement of a problem	uninterested, resigned	high fall, or rise-fall for concern
Fall on every yes/no question	impatient, interrogating	low rise
Narrow range throughout	bored, disengaged	wider excursions, higher onsets (7.5)
Flat `→Thank you` / `→Hello` / `→Sorry`	perfunctory, insincere	wide fall or fall-rise
Rise on everything	uncertain, needing reassurance	vary; fall on conclusions
Assertion where a hedge was expected	overconfident	fall-rise
No fall-rise available for "but…"	you appear to have no reservations, then surprise people with objections later	learn the fall-rise

The single highest-value item here is the fall-rise. It's the tune that lets you disagree, hedge, and deliver bad news while remaining pleasant.

Also: enthusiasm is prosodic in English and more lexical in French. [I] French conveys enthusiasm largely through word choice and intensifiers; English does it through pitch range. `That's great` with a narrow range means "fine, whatever". With a wide rise-fall it means what it says. If colleagues seem to think you're unenthusiastic about things you're actually keen on, this is why — and note the interaction with 7.5: L2 range narrowing means you may be *systematically* under-signalling enthusiasm across every conversation you have in English.

7.9 Melody drills

M1 — One tune, 15 sentences. Pick a tune (start with the fall-rise). Fifteen sentences, all with that tune, exaggerated. New tune next day. Cycle the six core tunes weekly.

M2 — Same sentence, six tunes. `You're going to Paris` with fall, low rise, high rise, fall-rise, rise-fall, level. Say what each means. Then with the short responses where tune does the most work per syllable: `Yes. No. Maybe. Okay. Sure. Right. Really. Fine. Great. Well.`

M3 — The tail drill. 7.1. Long tails after the nucleus: `MANagement, DIFficulty, INteresting, NECessary, COMfortable, posSIbility, deVELopment, TEMporary, CATegory`.

M4 — Humming the contour. Listen, hum it with no words, compare hums, then speak. The best single melody exercise, because it isolates pitch from articulation. Trace with your hand.

M5 — The `da-da-DA` transfer. Reduce a native sentence to nonsense syllables preserving rhythm and pitch: `də-də-DAA-də-də-\DA-də`. Say it. Then substitute the real words. No lexical content means all your attention goes to prosody.

M6 — Gesture mapping. A gesture per tune (hand drops for fall, rises for rise, dips-and-lifts for fall-rise, arcs for rise-fall). Motor coupling helps and builds a physical memory you can call on mid-conversation.

M7 — Compositional generation. New, and use it: pick a combination of tones from 7.2 that you haven't drilled — say `L* + L- + H%` — reason out what it should mean from the parts, produce it, then find a real example. This trains the system rather than the list.

M8 — Politeness rewriting. Ten blunt statements, softened prosodically (not lexically), then both:

```
That's wrong.          → That's \/not quite right.  → I'm 'not 'sure 'that's \/
right.
We can't do that.      → We 'can't \/really do that.
I disagree.            → \/Well… | I 'see it a 'little \/differently.
You didn't send it.    → I 'don't \/think I 'got it.
```

M9 — Emotional scales. One neutral sentence (`We're meeting on Thursday`) as: delighted, bored, suspicious, reassuring, irritated, surprised, doubtful, conspiratorial. Record; identify them blind later. If you can't, your prosodic range isn't carrying the distinctions yet.

M10 — The floor test. Say ten statements, each ending in a fall. In Praat, check the final F0 against your session minimum. Anything that stops more than ~3 semitones above your floor is an incomplete fall. Fix and repeat. This is the concrete replacement for "learn to creak."

PART VIII — ABOVE THE SENTENCE

Most pronunciation material stops at the sentence. Almost all of your professional English happens in longer stretches.

8.1 Paratones: the spoken paragraph

A **paratone** is a topic-sized prosodic unit. Its shape:

- **High onset** — the first accented syllable of a new topic starts noticeably higher than usual. A large pitch reset.

- **Gradual descent** — overall level and successive peak heights decline across it.
- **Low, long ending** — the final group ends very low, with long final lengthening, often creak, and a longer pause.

Immediately usable: - **New topic → start high.** Not louder. Higher. - **Finishing a point → go low and slow.** - **Continuing → moderate onset.**

In a presentation this is how listeners know where they are in your structure, and it works whether or not they're attending to your words. Delivering every sentence with the same onset height produces a structurally invisible talk: the audience can't distinguish a new section from a continuation, loses the outline, and disengages.

Drill. Take a five-paragraph text. Read it with a deliberately high onset at each paragraph start and a very low slow close at each end. Overdo it. Record and check you can hear the structure even with the words blurred.

8.2 Turn-taking and floor management

English signals turn transitions prosodically. If you don't produce the signals, natives will talk over you — a very common frustration for advanced non-native speakers, and almost always prosodic rather than social.

"I'm finished, your turn": pitch falls to the floor · final lengthening plus often creak · decreasing loudness · a completed syntactic unit.

"I'm not finished": non-low ending (fall-rise, high rise, or level) · **no** final lengthening — clip the last syllable · **keep the pitch up** across the boundary · a filled pause at a *non*-boundary · slight rate increase · audible in-breath *before* you finish the phrase, not after.

If you keep getting interrupted: you're almost certainly ending non-final groups too low with too much lengthening — producing the "I'm done" signal by accident. Practise:

`So the ˈfirst ˈthing is —(level, clipped)— and the ˈsecond ˈthing is…`

If you can never get in: start high and loud on the very first syllable. English turn-taking rewards a high-onset entry. `\SO —`, `ˈACtually —`, `Can I —` at a high pitch. Starting tentatively at mid-pitch guarantees you'll be talked over.

Backchannels. English listeners produce constant short signals: `mm-hm, uh-huh, right, yeah, okay, sure, I see, wow, exactly`. Each has its own tune, usually a small fall or a mid-level. Their absence makes you seem inattentive or unfriendly, especially on calls. French backchannels are less frequent and take different forms. Drill `mm-hm` (two syllables, second lower for acknowledgement, second higher for encouragement) and `uh-huh`.

8.3 Hesitation, fillers, thinking aloud

- **Place fillers at boundaries.** `The main problem — uh — is the timeline` ✓. `The main — uh — problem is…` ✗.
- **English fillers:** `um, uh, well, so, I mean, you know, like, sort of, let me think, how can I put this`. Note `um`/`uh` use `ʌ/ə`, not French `ø` — a `euh` in English is instantly identifying.

- **Lengthening as a filler.** English speakers stretch function words to buy time: `The problem is thəəə…`, `We need tooo…`, `and sooo…`. More native-sounding than a filled pause.
- **Discourse markers as time-buyers.** `Well`, `So`, `Actually`, `I think what happened was…`, `The thing is…`, `To be honest…`. Fill space, sound fluent, and buy a second or two of planning. Build a stock of five and drill them to automaticity — one of the fastest fluency wins available.
- **Self-correction prosody.** English speakers cut off sharply, then restart with a pitch reset and often a marker: `We sent it on Mon— actually on \TUESday.` The sharp cut and the reset are the signals; a smooth French-style continuation reads as confusion.

8.4 Parentheticals, asides, quotations

All marked by **register compression**: lower level, narrower range, faster rate, reduced loudness — then a return to normal.

```
The report — 'which ˌnobody ˌread — was 'actually 'quite \GOOD.
                (compressed, fast, low)

And 'then she said, (compressed) 'I'm not 'doing 'that, (return) and \LEFT.
```

Learners commonly give parentheticals the same prominence as the main clause, which makes the sentence structurally confusing. Practise deliberately dropping into a "smaller voice".

8.5 Meetings and presentations

- **Signposting with pitch.** `I have 'THREE \POINTS.` Then a high onset on each of `'FIRST`, `'SECond`, `'THIRD`. Numbered lists with flat onsets are useless; with high resets they're impossible to lose.
- **The rhetorical pause before the key word.** `And the result was … a 'complete \FAILure.` 400–700 ms. Effective and free.
- **Slow the nucleus, not the sentence.** Lengthen and widen the nucleus for emphasis; slowing the whole sentence just sounds laboured.
- **Contrastive structures.** English presentation rhetoric leans hard on prosodic contrast: `We 'didn't 'just \/MEET the target — | we \BEAT it.` `'This is 'not a \/COST — | it's an in'VESTment.` These need the fall-rise/fall pairing to work at all; flat, they're just sentences.
- **Numbers.** 3.11. Most meetings are half numbers, and number prosody is the most under-practised professional skill in this document.
- **Endings.** Close a section with a low, slow fall and a long pause. Close the talk the same way, more so. Trailing off on a rise reads as unfinished and produces a confused silence.
- **Video calls.** Compressed audio removes segmental information, so listeners rely *more* on prosody. Widen slightly, slow the nuclei, exaggerate boundaries, and pause longer than feels natural (latency eats your boundaries). Backchannel signals are muted or absent, so produce more verbal ones.

8.6 Reading aloud vs speaking

Two different prosodic registers, and a trap. Reading aloud tends to produce longer, more regular thought groups; fewer reductions and contractions; prominence on every content word; less register variation.

Practise only by reading aloud and you install *reading* prosody, then sound like you're reciting when you speak. Countermeasures: - Do read-aloud practice **after** hearing the passage read by a native, and imitate *their* phrasing rather than deriving it from punctuation. - Always finish a session with five minutes of **spontaneous** speech on the same topic. - Practise with **transcripts of unscripted speech** — podcast and interview transcripts — not written prose. The phrasing and reductions of spontaneous speech are what you need.

8.7 Voice setting — with appropriate caveats

Not intonation, but the substrate everything sits on. This area is described with more confidence than the evidence warrants, so the hedges below matter.

Every language is claimed to have a habitual **articulatory setting** — a default posture of lips, tongue, jaw and larynx between sounds. The classic descriptions (Honikman; Esling & Wong) characterise them roughly as:

- **French:** lips actively rounded and protruded much of the time; tongue tip forward and tense; tongue body relatively front; jaw fairly closed; even muscular tension.
- **General American:** lips more spread; jaw more open; tongue tip more retracted; tongue body raised and somewhat back; noticeable nasality; larynx a little lower; creak available as a normal register.

The caveats. [C] - These descriptions are **largely impressionistic**, and specific claims have been challenged. Esling & Wong describe American English as having "retroflex articulation" of the tongue tip, implying more retraction than French — but MRI work (Tiede, Boyce, Holland & Choe 2004) showed American `/r/` is articulated in a *variety* of ways, from retroflex to bunched, across and within speakers. - Laver's own framing acknowledges that no setting applies to every segment, and that a setting is audible only intermittently depending on how susceptible the current segment is. - Long-term average spectrum measures, sometimes offered as a "spectral signature" of a language, probably don't accurately describe the underlying setting.

What does hold up. [T] Work on French–English bilinguals (Wilson & Gick) supports the position that articulatory setting is a real, measurable component of being perceived as native, and pre-speech posture studies find native and near-native speakers rest close to their median speech position while lower-proficiency speakers have inefficient pre-speech postures.

So the practical version [I]: treat setting as a **global lever worth pulling**, not a precise specification to implement. Before an English session:

1. Relax and slightly drop the jaw — a little more open than feels French.
2. Un-round the lips. Neutral-to-spread, not protruded.
3. Pull the tongue body back and slightly up — the posture for `/ɚ/` in *bird*. Let that be your resting position. (See 2.3(2); this is why `/ɚ/` is worth learning early.)
4. Let the voice sit a bit lower and further back.
5. Don't fight nasality.

Say `Rrrrr… bird, word, sir, her, learn` and hold that posture. Then speak a paragraph from it. It will feel exaggerated. The value is that one adjustment shifts dozens of sounds at once without you fixing each individually — which matters because your attention budget is the scarce resource (10.1).

PART IX — LISTENING AND ANALYSIS

Given 1.6, this part is load-bearing for you. Budget at least a third of your practice time here, especially in the first months.

9.1 Why perception leads

1. **You can't store what you can't encode.** Every English word you learn without its stress pattern is a permanent liability. Improving perception improves the quality of every future word you acquire — it compounds.
2. **You can't self-monitor without perception.** Recording yourself is nearly useless if you can't hear the difference between your recording and the model. Many learners plateau exactly here: practising diligently against a standard they can't perceive, and consolidating errors.
3. **Perceptual training transfers to production. [E]** A meta-analysis of high-variability perceptual training studies found that perception gains carry over into production, across proficiency levels and target sounds. The reverse transfer is weaker.

9.2 The HVPT protocol

"Test yourself on minimal pairs" is roughly the right instinct, but it is unspecified to the point of being unusable. There is an evidence-based design.

What HVPT is. High Variability Phonetic Training: perceptual training in which learners are exposed to **multiple instantiations of the target contrast, produced by multiple talkers, in varied phonetic contexts, with trial-by-trial feedback**, usually via an identification or discrimination task.

How well it works. [E] - A meta-analysis of **79 studies** reports **medium-to-large effects**: $g \approx 0.92$ for pretest–posttest, $g \approx 0.67$ for treatment-vs-control. - **Long-term retention** of perception gains is confirmed. - There is **some generalisation to novel stimuli** — i.e. it isn't just memorising the training items. - Effects are moderated by length of L2 learning, response labels, task type, total training time, target phones, and number of talkers. - Typical immediate gains in the literature run around **10–15%** on target-sound perception. - It has been applied to **suprasegmentals**, including lexical tone and syllable structure. Its application specifically to *English word stress* is comparatively thin, so extending it there is an inference, not a demonstrated result. **[I]**

One caveat. [C] Whether *multiple* talkers actually beat a *single* talker is a genuinely mixed result across studies. Do use multiple talkers — the theory and most practice favour it — but don't treat single-talker practice as worthless.

Building it yourself

You need: a set of items, multiple voices, randomised presentation, immediate feedback, and a score.

Target contrast 1 — stress position in real words. 1. Take 60 words where you're unsure of the stress (from your 3.12 error list). 2. For each, obtain **three or four different speakers' recordings**. Forvo gives you multiple native recordings per word; YouGlish gives you the word inside real sentences from many speakers; a couple of different TTS voices can pad the set. 3. Build a two-alternative task: for each token, decide **which syllable is stressed** — or more simply, `first syllable / not first syllable`. 4. **Randomise.** Shuffle words *and* voices so you can't use talker-specific cues. 5. **Get feedback immediately after each item.** This is not optional; it's the defining feature. Studies without trial-by-trial feedback are excluded from HVPT meta-analyses. 6. **Score every session.** Log the number. 7. 15–20 minutes, most days. Total training time is one of the moderators, so accumulate hours.

Target contrast 2 — the reduced/full vowel judgement. Per 3.2, this is the cue English listeners actually weight most, so train it directly: for each token, judge whether syllable 2 (or 3) contains a **full vowel or a schwa**. Categorical, labelable, and directly relevant.

Target contrast 3 — nucleus location. Present short thought groups; identify which word carries the nucleus. Multiple speakers.

Target contrast 4 — tune identification. Six-way classification (fall / low rise / high rise / fall-rise / rise-fall / level). Hard; expect months.

Add perceptual fading. [T] For contrast 1, start with tokens where the duration difference between strong and weak syllables is exaggerated (you can stretch the stressed vowel in Audacity), then reduce the exaggeration as your score rises. This is the design that produced measurable improvement in trained French speakers on stress discrimination — fewer errors and faster responses than untrained controls.

A practical note. Building this by hand is tedious. Two shortcuts: (a) any flashcard app with audio can be repurposed — put the audio on the front, the stress pattern on the back, and it becomes an identification task with feedback; (b) if you're technical, an hour of scripting against a TTS API with several voices generates thousands of items and automates the scoring. Given your goals and the time you have available, that hour is probably the highest-leverage hour in this entire document. **[I]**

9.3 The perception ladder

Work up; each level is a prerequisite.

Level 1 — Can you hear that something is prominent? Count and tap the beats in a sentence. Don't identify words.

Level 2 — Can you locate it? `ba-BA-ba` vs `BA-ba-ba` — which syllable is strong. Start exaggerated, fade. Then real words, then unfamiliar words, then words in sentences.

Level 3 — Can you find the nucleus? Identify the most prominent word in a thought group; then say what it implies; then predict it before hearing.

Level 4 — Can you identify the tune? Six-way classification, isolated examples first, then running speech.

Level 5 — Can you transcribe a contour? Draw the pitch line over a transcript, marking accents, nuclei, boundaries and tunes. Then check against a Praat pitch track. Top of the ladder and the best ongoing practice.

Track numbers. Without scores you won't know whether perception is improving and you will assume it is. Score Levels 2–4 monthly.

9.4 Perception techniques

Humming / prosody-only listening. Hum a sentence back with no words. Strips lexical content, forces attention to pitch and rhythm. The best single perception exercise.

Low-pass filtering. Filter below ~400 Hz (Audacity: Effect → Filter Curve). Words become unintelligible; prosody survives. Do it with a native clip and your own clip of the same text and compare. Differences you couldn't hear become obvious. This is also Cutler & Butterfield's experimental method (1.3), and running it on yourself is unusually persuasive.

"Da-da" transcription. Write the rhythm as `də-də-DAA-də-\DA-də`. Forces explicit encoding.

Vowel-quality listening. Per 3.2, train on the cue English listeners use: listen to a word and judge only whether syllable *n* has a full vowel or a schwa. Ignore pitch.

Duration-only listening. Judge only which syllable is longest.

Function-word dictation. Transcribe fast native speech word for word, attending to function words. You'll miss `and`, `of`, `to`, `have` — evidence of how radically they're reduced, and calibration for your production.

Prosodic minimal pairs. Build sets and test blind: word stress (`'INsight / in'CITE`, `'DEsert / des'SERT`, `'PREsent / pre'SENT`); compound vs phrase (`'GREENhouse / green 'HOUSE`); nucleus (the six-way set); tune (`'isn't /it? / 'isn't \it.`); phrasing (the 5.4 set); numbers (`thir 'TEEN / 'THIRdy`).

ABX self-testing. A, B, and X — say which X matches. This is the actual paradigm from the stress-deafness research; a crude version is easy to rig in any audio editor.

9.5 Clip dissection

The core study activity. One clip properly dissected beats an hour of passive listening. 30–45 minutes per clip, 2–3 clips a week.

1. **Choose:** 20–40 seconds of *spontaneous* American speech from a speaker you'd be happy to sound like, on a topic near your professional register. Interview podcasts are ideal.
2. **Listen five times without transcribing.**
3. **Transcribe exactly** — fillers, false starts, contractions, reductions. Write `gonna`, not `going to`. This step alone is revealing.
4. **Mark boundaries.** Listen for final lengthening and pitch reset.
5. **Mark accents.** `'` on prenuclear, CAPS on each nucleus.
6. **Mark tunes.** `\ / \/ /\ →`.
7. **Mark reductions.** Circle every schwa, flap, deleted syllable, link.

8. **Open in Praat.** Check your markings against the pitch track. You will be wrong about things; that's the value.
9. **Note three surprises** in your notebook.
10. **Imitate:** chorus with the audio, then alone, then compare recordings.

After ten clips you'll have an empirical picture of American prosody no textbook can give you, and a calibrated ear.

9.6 Praat

Free, from the University of Amsterdam. Research on using it with learners consistently finds visual pitch feedback helps — precisely because it bypasses unreliable auditory self-perception, which for you is the point.

Setup: - `Open → Read from file`, then `View & Edit`. - `Pitch → Pitch settings`: **floor 60–75 Hz, ceiling 250–300 Hz** for your male voice. Wrong settings produce garbage — the default 75/600 will show octave errors. This is the number-one beginner mistake. - Turn on pitch (blue), intensity (yellow), spectrogram. - Select a stretch and use `Pitch → Get minimum/maximum pitch` for numbers.

What to look at:

Question	Where to look
Where are my accents?	pitch peaks — do they align with stressed syllables of content words?
Am I marking stress with duration?	select each syllable, read the duration; compute strong:weak ratio
Is my range narrowed in English?	min/max F0 in semitones, English **vs French** (7.5)
Is my melody at the edges or on the accents?	*where in the phrase* the biggest movements sit
Do I have declination?	draw a line over the peaks in a group — does it slope down?
Do I reset at boundaries?	is each group's first accent higher than the previous group's last?
Do my falls reach the floor?	final F0 vs session minimum (M10)
Am I reducing?	spectrogram: reduced vowels are short, formants pulled centrally

The killer comparison. Record yourself saying exactly what a native said. Open both, put the pitch tracks side by side. Differences you couldn't hear become impossible to miss. Weekly. The closest thing to a free coach.

Advanced: prosody transplantation. Praat can impose a native speaker's F0 contour and durations onto your own recording via PSOLA resynthesis. Hearing *your own voice* with native prosody is uncanny and instructive: it isolates prosody from everything else and demonstrates how much of your accent is prosodic. Published tutorials exist.

Supplements: Audacity (recording, A/B, tempo change without pitch change, low-pass filtering, stretching vowels for perceptual fading). Any real-time pitch tracker or tuner app, since Praat isn't real-time. Pronunciation apps only if they show a **pitch contour comparable to a model** — a score alone is useless to you.

9.7 The prosody notebook

One place, paper or digital. Sections:

1. **My stress errors** — words, with correct stress marked. Weekly spaced review.
2. **Clip dissections** — the marked-up transcripts.
3. **Surprises** — things that contradicted expectation.
4. **Tune log** — new uses noticed, with the example that taught you.
5. **My chunks** — phrases overlearned to native standard (10.2). Your assets.
6. **Measurements** — Praat numbers and perception scores, **dated**. How you know you're improving when it doesn't feel like it.

9.8 Models and material

Pick one primary model voice. One American speaker, same sex, similar age, register you want. Listen a lot, imitate specifically. Multiple models early dilutes the signal; broaden later. The single most efficient decision in the process.

Note the tension with 9.2: **perceptual training wants many voices; production imitation wants one.** These are different activities and the apparent contradiction is real but not a problem — vary the voices when training your ear, fix on one when training your mouth. **[I]**

Criteria for a model: General American · spontaneous, unscripted · clean recording · talks about what you talk about · available in quantity with transcripts.

Material, roughly by value:

1. **Interview podcasts with transcripts.** Spontaneous, professional register, long-form. Best single source.
2. **Conference talks, especially Q&A.** Q&A is unscripted professional English under pressure — exactly your target register.
3. **Sitcoms and dramas.** Excellent for tunes, attitude and conversational prosody; weaker on professional register. Short scenes repeated many times are gold for tune work.
4. **Audiobooks with the text.** Good for phrasing; watch the reading-register trap (8.6).
5. **Documentary narration.** Clear and wide-range, but a performed register.
6. **News.** Broadcast news has its own artificial prosody — very regular chunking, distinctive contours. Not a good general model.

On quantity. [I] Passive exposure has limited effect on adult production — you've had ten years of it. What produces change is attentive, analytic, repeated listening to small amounts. One minute analysed ten ways beats an hour in the background.

PART X — PRACTICE METHODOLOGY

You can put in serious time. How you spend it matters more than how much.

10.1 Motor-learning principles

Speech production is a motor skill with a perceptual control loop. These principles are well established in motor learning generally and apply here. Most are **[T]** or **[P]** as applied specifically to L2 speech.

1. Attention is the bottleneck, and it's tiny. You can consciously monitor roughly *one* prosodic dimension at a time. Try to attend to word stress, reduction, nucleus placement, tune and `/r/` at once and you'll do all of them badly. **One thing per session.** Name it before you start: "today, only vowel reduction." Everything else is allowed to be bad.

2. Blocked practice for acquisition, random practice for retention. Many repetitions of the same thing when learning a pattern — that's how you find the movement. Once you can produce it, **interleave**: mix it with other patterns in random order. Learners who only do blocked practice perform beautifully in drills and revert instantly in conversation. **This is the single most common reason pronunciation practice fails to transfer.**

3. Variability beats repetition. Same pattern in many different words, sentences, speeds and emotional colourings, rather than one sentence 100 times. (Note the convergence with HVPT — 9.2 — where variability is the defining feature on the perception side too.)

4. Feedback must be accurate, and delayed enough to let you self-evaluate first. Produce → *judge your own performance* → then check the model. Judging first is what trains perception. Checking immediately without self-judging makes you dependent on the model.

5. Spacing beats massing. 30 minutes daily beats 3.5 hours on Sunday. Motor consolidation appears to benefit from sleep, so daily practice with sleep between sessions does something a single long session can't.

6. Practise at the edge of ability. Comfortable means no longer training. Increase speed, reduce exaggeration, remove the script, add a distractor.

7. Fade the crutches deliberately. Every support — exaggeration, slow speed, written markings, listening immediately before speaking — must be planned to be removed. If it isn't removed, you've learned a supported skill.

8. Two systems, one goal. You have a *monitored* system (careful, slow, correct) and an *automatic* one (fast, spontaneous, habitual). Practice improves the monitored one first. The whole game is transfer, and it needs specific work (10.6). This is why people with excellent read-aloud pronunciation still have strong accents in conversation.

10.2 Imitation techniques

Chorusing — speak *simultaneously* with the recording. Loop a 3–8 second phrase, ten times round. Lowest cognitive load; best for installing rhythm and melody because the model carries you.

Shadowing — repeat with a lag of a few hundred milliseconds, following while the speaker continues. Harder: listening and speaking at once. **The technique with the most research behind it. [E]** A systematic review of 44 studies found shadowing improves comprehensibility, intelligibility, accentedness, and prosodic control (fluency, rhythm, intonation), while its effect on **individual segments was inconclusive**. That distribution is exactly what you want: shadowing is a *prosody* tool. Learners also consistently report finding it interesting and effective, which matters for adherence.

Two modes: - **Mumble shadowing** — hum or mumble the contour without articulating clearly. Excellent early stage; forces prosody, ignores segments. - **Full shadowing** — words and all.

Mirroring — shadowing plus the speaker's posture, gesture and facial expression. Sounds silly; works, because prosody is physically produced and copying the body copies breath support, jaw posture and gesture-pitch coupling.

Delayed imitation — listen, then repeat from memory. Higher load, closer to real production, better for consolidation. After chorusing.

Backchaining — build the phrase from the end backwards (R4). Preserves the nuclear tune and final rhythm, which is what runs out of steam building front-to-back.

Overlearning ("signature chunks"). Drill a short useful utterance to *automatic native-level* production — 50–100+ repetitions over several days, until it comes out perfectly without attention. Then bank it. Build a stock of 50–100.

Unglamorous and extremely effective, for two reasons. These are the things you say most often, so return per unit of practice is enormous. And an overlearned chunk requires **no attention**, which frees your attention budget for everything else — including, per 7.5, for pitch range. Fluent-sounding non-native speech is largely a matter of having a large stock of automatic chunks.

First ten chunks: your name and role · "nice to meet you" · "can you hear me okay?" · "let me share my screen" · "sorry, could you say that again?" · "I'm not sure I follow" · "that's a good question" · "what I mean is…" · "just to be clear" · "does that make sense?"

A 20-minute imitation session:

```
2 min   listen to the clip, no speaking
3 min   hum / mumble-shadow the contour
5 min   chorus, looped, phrase by phrase
5 min   delayed imitation, phrase by phrase, from memory
3 min   record yourself doing the whole clip
2 min   compare to the model — ONE dimension only
```

10.3 Exaggerate, then fade

The principle running through everything, and the one with direct empirical support in your specific case: the perceptual-fading study that improved French speakers' stress discrimination worked by **exaggerating the duration cue and then gradually reducing the exaggeration**. **[T]**

- Stretch stressed vowels to absurd lengths. `That's the MAAAAAIN prooooblem.`
- Swallow weak syllables until they nearly vanish.
- Make pitch movements operatic.
- Then dial down to 100% over weeks.

Why it works: at normal magnitude the contrast is below your perceptual and motor resolution. At 300% you can feel and hear what you're doing.

And the calibration error, which is real and worth verifying yourself. [I] Almost every learner who thinks they're exaggerating wildly is, on recording, merely correct. Record what feels like a ridiculous parody of an American and listen back; you will typically find it sounds like a mild, pleasant American accent. Verify this once and it will change how hard you're willing to push.

10.4 The physical toolkit

Rubber band. Stretch on stressed syllables, release on weak ones. Proprioceptive feedback on duration. Still the best tool for making duration tangible.

Conducting. Move your hand as if conducting — downbeats on accents, tracing the contour in the air. Big.

Metronome / tapping. 70–110 bpm for beat structure.

Humming and kazoo. Hum with lips closed, or use a kazoo, so articulation is impossible and only prosody remains. A kazoo genuinely helps: it makes your contour audible as pure melody.

Walking on the beats. Step on stressed syllables while reciting. Engages larger motor systems; good if hand gestures feel too small.

Jaw release. Chewing and stretching before practice, to release the tighter French posture. Combined with 8.7.

Singing. Songwriters align lexical stress to musical stress, so the music tells you the stress: `HAP-py BIRTH-day TO you`. Useful and pleasant.

Whispering. Whisper has no F0, so it isolates duration and intensity from pitch. Whisper a paragraph with correct rhythm, then voice it.

Dual-tasking. Speak while walking, washing dishes, light exercise. Deliberately stresses the automatic system and reveals what's actually automatic.

The `accent d'insistance` scaffold. 1.7. Say the English word as an emphatic French insistence, then subtract. Two-step, ten times.

10.5 The feedback loop

Non-negotiable, and given 1.6, more necessary for you than for most.

1. Record the model clip and your version.
2. **Before comparing, write down what you think you did wrong.** Predict. This trains perception.
3. Listen to yours, then the model. Alternate A/B in short segments.
4. Check **one** dimension. Rhythm today; melody tomorrow.
5. If you can't hear a difference, **look at the pitch tracks** (9.6). Visual feedback exists exactly for this.
6. Note the specific fix in one sentence. Redo.
7. **Keep dated recordings.** Monthly, listen to one from three months ago. Progress is invisible day to day and obvious over months — the best defence against quitting.

External feedback. You will need it, because self-perception is the weak link. - **A native speaker friend** willing to be blunt. Ask *targeted* questions ("did that sound like *deVELopment* or *developMENT*?"), never general ones ("how's my accent?"). - **A pronunciation coach**, even a few sessions. The highest-value money in this project. Ask specifically for someone who works on **prosody and suprasegmentals** — many coaches default to segments, which is not your bottleneck. - **Speech recognition as a crude proxy.** Dictate to a phone and see where it mis-transcribes. Systematic errors point at systematic problems. Blunt, free, unbiased. - **Blind listener tests.** Send a colleague two unlabelled recordings — yours now and yours from three months ago — and ask which is clearer.

10.6 Bridging to spontaneous speech

This is where nearly everyone fails. Correct prosody in imitation is common; correct prosody in unplanned speech is rare. The gap exists because imitation has no planning load — the model did the thinking. In spontaneous speech, lexical retrieval, grammar and content planning consume the attention prosody needs.

The solution is a **ladder**, each rung removing one support:

Rung	Task	What's given
1	Chorusing	everything; model present, simultaneous
2	Shadowing	everything; model present, slight lag
3	Delayed imitation	content; model absent, from memory
4	Marked-up reading aloud	content in text; prosody pre-planned by you
5	Unmarked reading aloud	content; prosody planned in real time
6	Retelling	content roughly; wording spontaneous
7	Prepared monologue	topic only
8	Semi-spontaneous, single focus	nothing; 2 min, one feature attended
9	Spontaneous + delayed review	nothing; analysed afterwards
10	Real conversation, one in-flight focus	nothing

How to work it: don't climb it once. Every new feature starts at Rung 1 and must be walked up. And practise at two or three rungs simultaneously — Rungs 1–3 for what's new, Rungs 7–10 for what's consolidating.

Specific bridging exercises:

- **The one-feature day.** Pick a feature. For a whole day, in all your real English, attend only to that. Accept that everything else regresses.
- **The 2-minute daily monologue.** Record yourself on a random prompt. Every day, same time. Review only 30 seconds, checking one thing. Over months this becomes your best progress record and your main transfer exercise.
- **Pre-planning real utterances.** Before a meeting, decide the prosody of your three key sentences. Mark them up. Then say them. This is how professional speakers work; it's entirely legitimate.
- **Self-repair practice.** In conversation, when you catch a stress error, repair it out loud — "sorry, *deVELopment*." Repairs are practice at the ideal moment, socially completely normal, and they hook the correction to a real memory.

- **Slow-motion spontaneous speech.** Talk spontaneously at half speed with big prosody. Absurd but effective: it gives your planning system enough slack to also handle prosody. Then speed up.

10.7 The focus menu

This is a menu, not a schedule. Pick one. Work it until it's boring, which might be two days or three weeks. Then pick another. Revisit anything at a higher standard. There is no order and no schedule.

Focus	Sections	Rough size
The four segmental prerequisites	2.3	small, do early
Vowel reduction: make weak syllables genuinely weak	3.10, 4.2	large
Word stress on your professional vocabulary	3.9, 3.12	large, ongoing
Long-word rhythm and secondary stress	3.3	medium
Weak forms and contractions	4.3	medium
Flapping and `/nt/` reduction	4.4	small
Numbers, dates, initialisms	3.11	small, high payoff
Thought-group boundaries: lengthening and reset	5.2	medium
The collision drill (stress at boundaries)	5.6	small, high payoff
Nucleus placement	6.2, 6.3	large
Deaccenting given information	6.6	medium
Falls that reach the floor	7.3 tune 1, M10	small
The fall-rise	7.3 tune 4, G6	medium, high payoff
Wh-question falls	7.3 tune 2	small, quick win
Pitch range: removing the suppression	7.5	ongoing
Downstep for presentations	6.5	medium
Turn-taking cues	8.2	small
Voice setting	8.7	small, do early
Perceptual training (runs in parallel, always)	9.2	permanent

The last row isn't a focus; it's a constant. Everything else rotates around it.

10.8 Common practice mistakes

Mistake	Why it fails	Instead
Practising words in isolation	doesn't transfer; isolated words carry citation prosody	carrier phrases and real sentences
Many features at once	attention overload	one per session
Only reading aloud	installs reading prosody	finish with spontaneous speech
Slowing down uniformly	flattens the contrast clarity depends on	increase contrast, keep weak syllables short
Not recording	you cannot perceive your own errors	record everything
Comparing without predicting	trains dependence, not perception	predict first
Never exaggerating	contrasts stay below your resolution	300%, then fade
Exaggerating forever	never reach natural	plan the fade
Only blocked practice	drill-perfect, conversation-hopeless	interleave once you can produce it
Multiple models early (for production)	dilutes the target	one primary model — but many voices for perception (9.8)
Only production work	your bottleneck is perception	≥⅓ on listening
Bursts	no consolidation	daily, spaced
Passive listening as "practice"	ten years hasn't fixed it	analytic listening on small samples
No measurement	you can't tell if it's working, so you quit	perception scores, Praat numbers, dated recordings
Chasing native-like segments before prosody	wrong order for your stated goals	prosody first

10.9 What to deprioritise

A document this long implicitly says everything matters. It doesn't. Given your stated goals — clarity first, correctness second, accent reduction third and not needing to be perfect — here is what to *not* spend much time on. **[I], and reasonable people would disagree with some of it.**

Deprioritise heavily:

- **Creak / vocal fry.** 7.6. Recognise it; don't practise it.

- **Compound stress rules.** 3.7. The literature says they're unreliable, and getting one wrong costs little because both content words survive intact. Verify your fifty most-used compounds and move on.
- **The full tune inventory.** Six tunes, well controlled, beats twelve half-controlled. The research on whether all twelve are genuinely distinct categories is equivocal anyway.
- **ToBI transcription fluency.** Useful to *read*. Learning to *produce* accurate ToBI labels is a research skill, not a speaking skill.
- **Regional American varieties.** Pick General American and ignore the rest until you have a specific reason.
- **Uptalk.** Worth being able to produce, not worth drilling. And it carries an assertiveness cost you don't need.
- **Perfecting `/r/` and `/θ/`.** These are accentedness features, not intelligibility features. A French `/ʁ/` in a rhythmically-native sentence is charming; a perfect American `/r/` in an evenly-timed sentence is not. (The exception is `/ɚ/` in weak syllables — 2.3(2) — which *is* prosodic infrastructure.)
- **Vowel inventory precision.** Same reasoning, with one exception: you need a real schwa, and you need `/ɪ/` distinct from `/iː/` in weak syllables. The rest can wait.
- **Idiom and slang acquisition.** Not a pronunciation problem, and at C1 you have enough.
- **Anything that produces a score without showing you a contour.** Most pronunciation apps.

Actively avoid:

- **Speaking slowly to be clearer.** 4.6. Counterproductive.
- **Over-articulating full forms.** 3.10, 4.3. Makes you *less* intelligible, not more.
- **Suppressing prosody to avoid sounding sing-song.** 7.7(7). Trades one problem for a worse one.
- **Practising only what's already comfortable.** 10.1(6).

The honest summary [I]: roughly 70% of your available gain is in reduction, word stress, and nucleus placement. Another 20% is in boundaries, the fall-rise, and range. The remaining 10% is everything else in this document, and it costs more than the first 90% combined.

10.10 The identity problem

Worth naming, because it stops people.

Imitating another accent feels like pretending. Producing a wide American pitch range feels theatrical or unserious to a French sensibility. Many learners unconsciously hold back because doing it properly feels like a betrayal of self or an embarrassing performance. This is well documented in pronunciation learning and it's the invisible ceiling on a lot of otherwise diligent practice.

Note that this connects directly to 7.5: the L2 range narrowing found in French and German speakers of English was attributed partly to **reduced confidence**. Some of your prosodic suppression may be inhibition rather than incapacity, which means some of the work is psychological.

What helps:

- **Frame it as a role, not a replacement.** You're adding a register, not deleting your French self. Bilinguals routinely have different voices, gestures and even personality expression in each language. Normal, not fake.
- **Practise as a character first.** Do a deliberately over-the-top American impression, privately, for fun. Let it be a performance. The physical skills transfer, and performance mode carries no self-consciousness.
- **Verify the calibration error.** 10.3. Each time you record yourself "wildly exaggerating" and hear something merely correct, the inhibition weakens.
- **Accept a target that isn't native.** Your stated goal already does. Keeping an audible French accent while being perfectly clear and prosodically native-like is a completely respectable and attainable target — arguably the best one available, and much healthier psychologically than "pass for American."
- **Volume of private practice** is what makes the public version unremarkable.

10.11 Troubleshooting

Symptom	Likely cause	Action
"Fine in drills, not in conversation"	no work on Rungs 6–10; only blocked practice	interleave; one-feature days; daily monologue
"I can't hear any difference between me and the model"	perception ceiling (1.6)	Praat visuals; HVPT with fading (9.2); external feedback
"I've stopped improving"	practising what's already automatic	raise difficulty: faster, less exaggeration, no script, dual-task
"People still ask me to repeat things"	almost always word stress or nucleus placement, not vowels	audit stress on your vocabulary; check nucleus placement
"People say I sound cold or aggressive"	missing fall-rise, narrowed range, bald falls	7.8
"I get interrupted constantly"	ending non-final groups too low	8.2
"I sound robotic / too careful"	over-monitoring, no reductions, reading prosody	more spontaneous work, force contractions, accept errors
"I sound sing-song"	melody at edges rather than on accents; continuation rises everywhere	7.1 tail drill; vary non-final tunes
"Fine slow, falls apart fast"	contrast produced by extra time rather than by compression	metronome drill; compress weak syllables instead of expanding strong ones
"My range is flat however hard I try"	it's load, not capacity (7.5)	overlearn more chunks to free attention; measure your French range for comparison
"It regressed after two weeks off"	normal motor-skill decay	short daily maintenance beats long absences
"Speaking English exhausts me"	over-monitoring	overlearn more chunks; accept more errors

On timelines [P] — these are practitioner estimates, not research findings. Noticeable improvement in *self-perception* within weeks. In *monitored* production, 1–3 months. In *spontaneous* production, 4–8 months of consistent daily work. Substantial change in how you're perceived at 6–18 months. **Word stress on your core professional vocabulary is much faster than any of that** — it's knowledge, and a few hundred words is a few weeks. Start there for the morale.

PART XI — THE EXERCISE LIBRARY

Small, self-contained. Pick from these to build sessions. Nothing here needs more than 15 minutes; most need 3–5.

A. Warm-ups (3–5 min before any session)

A1. Setting reset (2 min). Jaw stretches, lip release, then hold the `/ɚ/` posture and hum. Then `bird, word, heard, learn, turn, sir, her, third, work` slowly. Establishes the American setting (8.7).

A2. Range sweep (2 min). Hum bottom to top of your range and back, five times. Then on `mmm-AAAH`. Then `Hello` at five different heights. Opens the range you'll need (7.5).

A3. Schwa machine (2 min). `ə-ə-ə-ə-ə` fast and light, checking in a mirror that your lips don't round (2.3). Then `bə-də-gə-pə-tə-kə` at speed. Then the weak forms `THE, A, TO, OF, FOR, AND, CAN, WAS, HAVE` in weak form only, fast, ten times through.

A4. Floor drill (1 min). Say `Ahhh` letting pitch fall to the very bottom. Five times. Then `\NO. \YEAH. \RIGHT.` each falling all the way. Installs the American final fall — and creak may appear as a by-product, which is fine (7.6).

A5. Flap sprint (2 min). `water, better, city, later, matter, letter, pretty, little, thirty, party, notice, computer, whatever, get it, put it, lot of, not at all, out of it.` Fast, ten times.

A6. Syllabic consonant sprint (1 min). `button, bottle, mountain, sudden, written, little, middle, travel, total, national, certain, personal` — never opening the mouth after the closure (2.3).

B. Perception — the priority set

B1. Beat counting (3 min). Play a sentence, count and tap the beats, check against the transcript. Ten sentences.

B2. Which syllable (5 min). HVPT-style, per 9.2: multiple voices, randomised, immediate feedback, scored. Start with exaggerated duration, fade over weeks. **The most targeted exercise for your L1-specific deficit.**

B3. Full-or-schwa (5 min). Same format, but judge whether syllable *n* has a full vowel or a schwa. Trains the cue English listeners actually weight most (3.2).

B4. Blind stress guessing (5 min). Twenty unfamiliar polysyllabic words. Guess the stress before listening. Score. Track monthly — this number is a direct measure of progress on the core problem.

B5. Hum-back (5 min). Listen to a sentence, hum it back, record, compare hums not words.

B6. Low-pass listening (5 min). Filter below ~400 Hz. Describe the melody in words. Then unfilter and check.

B7. Nucleus spotting (5 min). Ten thought groups; name the most prominent word; say what it implies.

B8. Tune identification (5 min). Twenty short utterances, six-way classification. Score.

B9. Function-word dictation (5 min). Transcribe 30 seconds of fast native speech word for word. Count what you missed.

B10. Minimal-pair blind test (5 min). Any set from 9.4, randomised, scored.

B11. Prediction (5 min). Mark where *you* think nuclei and boundaries go on a transcript line, then listen. Errors here are the most informative data available.

B12. The filtered-rhythm illusion (once, 5 min). Low-pass filter `conduct ascents uphill` (record it yourself) and see whether you hear *the doctor sends the bill* (1.3). Worth doing once, for the conviction it produces.

C. Word stress

C1. Family recitation (3 min). A stress-shift family (3.6) in a row, tapping beats. Five families.

C2. Da-da mapping (3 min). Ten long words: say the `da-də` rhythm three times, then the word.

C3. Carrier-phrase drill (5 min). Twenty error-list words in `The ______ was the issue.` / `We need to discuss the ______.`

C4. Noun/verb pairs in sentences (5 min). `We need to record the record.` / `They'll present the present.` / `The increase will increase costs.` / `Don't object to the object.`

C5. `-ate` pairs (3 min). `a separate room / to separate them`, `an estimate / to estimate`, `a graduate / to graduate`, `appropriate / to appropriate`, `a delegate / to delegate`, `moderate / to moderate`.

C6. The `-ary`/`-ory` sprint (3 min). `necessary, secretary, temporary, ordinary, category, territory, dictionary, military, inventory, laboratory, cemetery, February, ceremony`. Fast, ten times, with the reduction. Where French speakers lose the most.

C7. The insistence scaffold (5 min). 1.7, two-step. Twenty initially-stressed words: French emphatic insistence version, then English target version, alternating.

C8. Number pairs (2 min). `13/30 … 19/90` in real sentences: `We need thirty, not thirteen. Fourteen fifty, not forty fifty.`

C9. Initialism stress (2 min). `the U-S-\A, C-\EO, A-P-\I, K-P-\I, H-\R, C-R-\M, U-\X, S-Q-\L` — stress on the last letter (3.11).

C10. Cognate hunt (5 min). A page of your own professional writing. Circle every French-looking word. Check its English stress. Add errors to the notebook.

C11. Overlearn ten (5 min). Your ten worst words, twenty repetitions each in a sentence, daily for a week.

D. Rhythm and reduction

D1. Metronome fit (5 min). 90 bpm, three sentences, ten repetitions each. Then 70 and 110.

D2. Same-beats expansion (3 min). R2: `DOGS EAT BONES` → `The dogs would have eaten all of the bones` in the same time.

D3. Limerick (3 min). A memorised limerick at increasing speed, five times.

D4. Backchaining (5 min). Three long sentences built from the end.

D5. Weak-form respelling (5 min). Rewrite a paragraph phonetically with all reductions; read your respelling.

D6. Syllable deletion (3 min). `comfortable, vegetable, chocolate, family, every, camera, different, business, average, several, general, temperature, probably, interesting, restaurant, laboratory, February` at the *reduced* syllable count. Ten times.

D7. Contraction forcing (3 min). Read a paragraph converting every possible full form to a contraction, then again with multi-word reductions.

D8. Whisper rhythm (3 min). Whisper a paragraph with correct rhythm (no pitch available), then voice it.

D9. Linking sprint (3 min). `pick it up, turn it off, an apple, a lot of it, find out, take it away, get out of here, put it on, look at it, one of them, not at all.` Fast, linked, ten times.

D10. Rubber band (5 min). Any sentence set, stretching on stressed syllables.

D11. Unreleased finals (2 min). `stop, that, back, bad, hot, wait, took` — freeze on the closure, never open. Then in phrases: `Let's stop. I want that. Sit back.`

E. Phrasing

E1. Mark-up-then-read (5 min). Mark boundaries and nuclei on a paragraph, then read it.

E2. Boundary-only transcription (5 min). Listen and mark only boundaries; compare with your instinct.

E3. One-breath-per-group (3 min).

E4. Ambiguity pairs (5 min). The 5.4 set, both members.

E5. Same sentence, N phrasings (3 min).

E6. The collision drill (5 min). 5.6. Non-final-stressed words at boundaries and at the end of rising questions. `Is it imPORtant?` `Did you check the comPUter?` `Is that the deVELopment team?` **Highest-value phrasing drill for a French speaker.**

E7. Number-string chunking (3 min). Phone numbers, account numbers, dates — rise on each group, fall on the last (3.11).

F. Prominence

F1. Six-way sentence (5 min). All placements, plus the implicature written in French.

F2. Q→A nucleus (5 min). Ten pairs; nucleus on new information, given deaccented.

F3. Contrast pairs (3 min). Ten `not X but Y` structures.

F4. Deaccenting dialogue (5 min). Read a real transcript deaccenting all given information.

F5. Nucleus-only shadowing (3 min). `DA` on each nucleus, silent elsewhere.

F6. Number contrasts (3 min). `from 'THIRty to 'FORty`, `Q\THREE not Q\TWO`, `'THIRteen, not 'THIRdy`. The most frequent professional use of contrastive accent.

F7. Function-word contrast (3 min). `It's ON the table, not UNDER it.` `I DID send it.` `It's not MY problem, it's YOURS.`

F8. The nucleus-exception set (5 min). Work 6.2's exceptions (b) through (i) — empty nouns, infinitives, adjectives after nouns, `what KIND of`, relative clauses, phrasal verbs, event sentences. Five examples each.

F9. The "so what" habit (all day). After each sentence you say, silently ask which word was the point, and whether the nucleus went there.

G. Melody

G1. One tune, 15 sentences (5 min).

G2. Six tunes, one sentence (5 min). Especially the one-word responses: `Yes. No. Maybe. Okay. Sure. Right. Really. Fine. Great. Well.` Most meaning per syllable of anything in English.

G3. The tail drill (3 min). 7.1. **Highest-value melody drill for a French speaker.**

G4. Humming a contour (5 min).

G5. Gesture mapping (3 min).

G6. Fall-rise immersion (5 min). Twenty fall-rise utterances. `I \/LIKE it. It's \/POSSible. \/SOME of them. We could \/TRY. I \/THINK so. \/Well… As far as \/I know. \/Maybe. It's 'not \/impossible. I'm 'not \/sure.` **Spend a week on this tune.**

G7. Politeness rewriting (5 min). Ten blunt statements, prosodically softened.

G8. Emotional scale (5 min). One sentence, eight emotions, recorded, blind-tested later.

G9. Wh-question falls (3 min). Fifteen wh-questions, all falling. `Where are you \GOing? What's the \PROBlem? Why did you \DO that? How much does it \COST? When can you \SEND it?` Then contrast with echo-rise versions.

G10. Downstep staircase (3 min). `The 'FIRST 'thing | the 'SEcond 'thing | and \FInally…` each peak lower.

G11. Compositional generation (5 min). Pick an untried tone combination from 7.2, reason out what it should mean from the parts, produce it, then find a real example. Trains the system rather than the list.

G12. The floor test (3 min). Ten falling statements; check final F0 against your session minimum in Praat. Anything stopping more than ~3 semitones above the floor is incomplete.

G13. Range comparison (3 min). Read the same content in French and in English. Measure both spans. The gap is your suppression, quantified (7.5).

H. Discourse and integration

H1. Paratone reading (5 min). Five paragraphs, high onsets and low closes.

H2. Turn-holding drill (3 min). Five non-final chunks with the "don't interrupt" signature: level ending, clipped last syllable, pitch up.

H3. Backchannel drill (2 min). `mm-hm, uh-huh, right, yeah, sure, okay, I see, exactly, wow` with appropriate tunes.

H4. Filler placement (3 min). Speak for a minute with deliberate fillers at boundaries only, using ʌ/ə not ø.

H5. Parenthetical compression (3 min). Five sentences with asides, dropping into a compressed register.

H6. Signposted mini-presentation (5 min). Ninety seconds, three signposted points, high onsets, low closes. Recorded.

H7. The 2-minute monologue (3 min). Random topic, recorded, daily. Review 30 seconds for one feature.

H8. Self-repair practice (in real life).

H9. One-feature day (all day).

H10. Full clip dissection (30–45 min, 2–3× per week). The 9.5 protocol. Your main study activity.

H11. Your own name (once, then bank it). 3.11. Decide your version, overlearn it, never improvise it again.

PART XII — REFERENCE

Appendix A — Notation

Symbol	Meaning
`ˈ`	primary stress, before the syllable
`ˌ`	secondary stress
`CAPS`	nucleus of the thought group
`\`	fall
`/`	rise
`\/`	fall-rise
`/\`	rise-fall
`→`	level
single / double pipe	minor / major boundary
`ə`	schwa
`ɚ`	r-coloured schwa
`‿`	linking
`H* L*`	high / low pitch accent (ToBI)
`L+H*`	rising pitch accent — contrastive
`L*+H`	low accent then rise — uncertainty
`!H*`	downstepped high accent
`H- L-`	phrase accent
`H% L%`	boundary tone
break indices	`0` fully joined · `1` word boundary · `3` intermediate phrase · `4` full phrase

Appendix B — Weak forms

`a` ə · `an` ən · `the` ðə/ði · `and` ən/n̩ · `but` bət · `or` ɚ · `as` əz · `than` ðən · `that`(conj) ðət · `of` əv/ə · `to` tə/də · `for` fɚ · `from` frəm · `at` ət · `can` kən/kn̩ · `could` kəd · `will` l̩ · `would` d/wəd · `should` ʃəd · `must` məs · `have` (h)əv/v · `has` (h)əz/z · `had` (h)əd/d · `am` əm · `are` ɚ · `was` wəz · `were` wɚ · `be` bi · `been` bən · `do` də · `does` dəz · `he` i · `him` ɪm · `his` ɪz · `her` ɚ · `them` ðəm/əm · `us` əs/s · `you` jə · `your` jɚ · `some` səm · `there`(exist.) ðɚ · `just` dʒəs

Strong forms return when: contrastively stressed · phrase-final · cited.

Appendix C — Nucleus placement

1. **Default:** on or near the last word of the group *that matters for the meaning.*

2. **Skip** final time/place adverbials unless they're the point.
3. **Skip** empty nouns (`thing, people, one, place, stuff`).
4. **Skip** pronouns and indefinites.
5. **Keep it on the noun** before an infinitive (`a \LETter to write`), before an adjective/participle (`the only \OPtion available`), and as a relative-clause head (`the \FILE you asked for`).
6. **In `what KIND of X`** the nucleus precedes the light noun.
7. **Transitive phrasal verbs:** nucleus on the object (`turn the \LIGHT off`).
8. **Event/intransitive sentences:** subject-nucleus is available and common (`The \PHONE's ringing`) but genuinely variable — not a rule.
9. **Move it** for contrast — any word, including function words.
10. **Never** on given information. Given = deaccented.
11. **Exactly one** per thought group.

Appendix D — Suffix stress

Neutral (stress doesn't move): -ing, -ed, -s, -er, -est, -ly, -ness, -ment, -ful, -less, -able, -ish, -hood, -ship, -dom, -ism, -ize, -age *(but watch the American shift in `ˌneces'sarily, ˌtempo'rarily, ˌvolu'ntarily`)*

Auto-stressed (suffix takes it): -ee, -eer, -ese, -esque, -ette, -ique, -oon, -aire, -ade, -self

Stress immediately before the suffix: -ic(al), -ity/-ety, -ion (+ -ian, -ial, -ious, -eous, -ual, -uous, -ium, -iar, -ience), -ify, -itis, -osis, -escence

Stress two syllables before: -ate (verbs), -graphy/-grapher, -logy, -cracy, -meter, -itude, -itute

Pushed further left: C + -ary, -ory, -ery, -ony

Word class: 2-syllable nouns/adjectives → first · 2-syllable verbs → second

Compounds: noun+noun → usually left, **but** material / temporal / locative / created-by / copulative relations favour right; lexicalisation, frequency and one-word spelling favour left. Verify rather than derive.

Wells's generalisation: every English word has a lexical stress on one of its first two syllables.

Appendix E — Tunes

Shape	ToBI	Composition	Use
\	`H* L-L%`	new + self-contained + mine	statements, wh-questions, commands, finality. Reach the floor.
/ from low	`L* H-H%`	not-asserted + forward + yours	yes/no questions, checking, politeness
/ from high	`H* H-H%`	asserted + forward + attend	continuation, turn-holding, narrative
\/	`L+H* L-H%`	contrastive + self-contained + yours	reservation, "but…", hedging, politeness. **Learn this one.**
/\	`L+H* L-L%`	contrastive + self-contained + mine	emphasis, contrast, surprise, enthusiasm
→	`H* H-L%`	—	routine, list items, non-committal
staircase	`!H*` series	—	calm, structured declaratives
low + final rise	`L*…L*+H L-H%`	not-asserted + yours	contradiction: "you've got that wrong"
step-down level	`H* !H- L%`	—	calling, chanting, announcing
all low	`L* L-L%`	—	bored, grim, dismissive

Element meanings (7.2): `H*` = new/asserted · `L*` = salient but not asserted · `L+H*` = contrastive/scalar · `L*+H` = contrastive but uncommitted · `H+L*` = should have been inferable · `H-` = forward-looking · `L-` = self-contained · `H%` = burden to the hearer · `L%` = burden mine.

Appendix F — 30-item self-diagnostic

Score 0 (never) to 3 (consistently). Re-take monthly. Anything at 0–1 is a target.

Segmental prerequisites 1. My schwa is unrounded and central, not `/ø/` or `/œ/`. 2. I produce `/ɚ/` without French uvular friction. 3. I produce syllabic n̩ and l̩ without inserting a vowel. 4. I don't release final stops with a trailing vowel.

Word level 5. I know the stress of the 200 words I use most at work. 6. My unstressed syllables have schwa, not full vowels. 7. My strong:weak duration ratio is at least 3:1. 8. I produce secondary stress in long words. 9. I say `comfortable/vegetable/every/different` at the reduced syllable count. 10. I distinguish `-ate` verbs from `-ate` nouns/adjectives. 11. I get `thirteen/thirty` right, both ways, in context. 12. I know when a French-looking word has earlier English stress. 13. I stress the last letter of initialisms.

Rhythm 14. I use weak forms of function words. 15. I use contractions automatically in speech. 16. I flap intervocalic `/t/`. 17. I reduce `/nt/` in `twenty, internet, center`. 18. I link consonants to following vowels. 19. My syllables are audibly unequal in length. 20. I can fit a 12-syllable and a 3-syllable phrase into similar time.

Phrasing 21. I chunk into 2–7-word groups. 22. I never pause between a preposition and its object. 23. I mark boundaries with final lengthening. 24. I reset pitch upward at each new group. 25. I keep word stress correct at a boundary and in a rising question.

Prominence and melody 26. Exactly one nucleus per thought group. 27. I put it on the informationally important word. 28. I deaccent given information. 29. My falls reach the bottom of my range. 30. I use falls, not rises, on wh-questions — and I have and use the fall-rise.

Appendix G — Myths worth discarding

1. **"English is stress-timed with equal inter-stress intervals."** False as stated. The useful truth: English compresses weak syllables far more than French.
2. **"Speaking more slowly makes you clearer."** Usually the reverse; it flattens the contrast clarity depends on.
3. **"Full pronunciation is careful pronunciation."** No. Four-syllable `comfortable` is wrong, not careful.
4. **"Reductions are lazy or informal."** They are obligatory features of all registers of native English.
5. **"Uptalk means uncertainty."** Mostly signals continuation, turn-holding and engagement-checking. It does carry an assertiveness cost in some contexts — a reason to use it selectively, not to believe the folk explanation.
6. **"Intonation is emotional decoration."** It's grammar and information structure. Wrong intonation changes what you said.
7. **"You just need more exposure."** Ten years hasn't done it. Adults need explicit, analytic, feedback-driven practice.
8. **"You can't improve pronunciation after childhood."** Comprehensibility and prosody are demonstrably trainable in adults. Indistinguishable-from-native is rare; fully clear with near-native prosody is not.
9. **"Work on sounds first, then prosody."** Backwards for your goals — with the narrow exception of the four prerequisites in 2.3.
10. **"Wh-questions rise."** They fall.
11. **"If I can hear it, I can learn to say it."** Also watch the reverse: given 1.6 you may not be able to hear something you nonetheless need to produce. Use visual feedback.
12. **"Louder = more stressed."** Vowel quality and length do most of the work; loudness the least.
13. **"French has a narrow pitch range, so widen yours."** No good evidence French is narrow. Your *L2* range is narrowed. Different diagnosis, different fix (7.5).

14. **“Americans creak, so learn to creak.”** Variation between individual American speakers is larger than any French/English difference found. Aim at reaching the floor; creak is a by-product (7.6).
15. **“Noun+noun compounds are left-stressed.”** Called a myth in the literature for good reason (3.7).

Appendix H — Resources

Tools - **Praat** (free, praat.org) — pitch tracks, durations, spectrograms, prosody transplantation. The core tool. Set the floor/ceiling correctly (9.6). - **Audacity** (free) — recording, A/B, tempo change without pitch change, low-pass filtering, vowel stretching for perceptual fading. - **A metronome app.** - **Forvo** — multiple native recordings per word. Your source of talker variability for HVPT (9.2). - **YouGlish** — a word inside hundreds of real video clips. Best tool for hearing a word inside real prosody. - **Merriam-Webster / Cambridge (US)** — reliable US audio and stress marking. - **A flashcard app with audio** — repurposed as an HVPT identification task. - **A pronunciation/accent coach specialising in prosody** — the best money in this project.

Books - **J.C. Wells, *English Intonation: An Introduction*** — the best practical treatment of tunes and nucleus placement, with audio. British-focused; the framework transfers directly. Chapter 3 on tonicity is the single most useful chapter for your Fault B. - **Peter Roach, *English Phonetics and Phonology*** — clearest general introduction. - **Judy Gilbert, *Clear Speech*** — prosody-first, pedagogically excellent, designed for exactly your problem. - **Linda Grant, *Well Said*** — advanced American pronunciation, strong on suprasegmentals. - **Ann Cook, *American Accent Training*** — very practical on rhythm and American “staircase” intonation. Somewhat mechanical; effective. - **Collins & Mees, *Practical Phonetics and Phonology*** — good reference with L1-specific sections. - **Geoff Lindsey, *English After RP*** — how modern English actually sounds versus textbook descriptions. His online material is excellent. - **Alan Cruttenden, *Intonation*** — the scholarly reference if you want depth. - **D.R. Ladd, *Intonational Phonology*** — for the theory behind Part 2.5 and 7.2.

Note: software and apps date quickly; verify what’s current. The selection criterion is always *does it show me a pitch contour I can compare against a model*, not *does it give me a score.*

Appendix I — Glossary

Accent (prosodic) — prominence given to a syllable in an utterance, realised with pitch movement. Distinct from *stress* (a lexical property) and from *accent* (a way of speaking). **Accentual phrase (AP)** — the French prosodic unit; underlying tonal shape /LHiLH/. ***Boundary tone** — the pitch target at the edge of an intonation phrase. **Broad focus** — no particular contrast; default nucleus placement. **Deaccenting** — removing prominence from given information. **Declination** — gradual downward drift of pitch across a phrase. **Downstep** — each successive accent peak lower than the last. **Eurhythmy** — the preference for alternating strong and weak beats. **Flapping** — intervocalic `/t/` → a quick voiced tap. American. **Foot** — a stressed syllable plus following unstressed ones. **Glottalisation** — irregular phonation; clusters at prosodic boundaries and accents. **Head** — the stretch from the first accent to just before the nucleus. **Hi** — the French initial/secondary accent; a left-edge phrase accent. **HVPT** — High Variability Phonetic Training. **Intonation phrase** — the* *tune-bearing unit; = thought group.* ***Metrical Segmentation Strategy*** *— English listeners' use of strong-syllable onsets as word-boundary hypotheses.* ***Narrow focus*** *— contrastive or emphatic nucleus placement.* ***Nucleus*** *— the last accented syllable in a phrase; carries the nuclear tune.* ***Paratone*** *— a spoken "paragraph".* ***Perceptual fading*** *— training design that exaggerates a cue then reduces the exaggeration.* ***Pitch accent*** *— a tonal target anchored to a stressed syllable.* ***Pitch reset*** *— the upward jump at the start of a new phrase.* ***Prosody*** *— all suprasegmental properties of speech.* ***Reduction*** *— shortening and centralising of unstressed vowels.* ***Schwa*** *— the reduced central vowel ə.* ***Stress "deafness"*** *— French speakers' documented difficulty encoding stress* lexically*; see 1.6 for the qualified version. **Suprasegmental** — pertaining to units larger than the individual sound. **Tail** — the unaccented material after the nucleus. **ToBI** — Tones and Break Indices; the standard American English intonation transcription system. **Tune** — the pitch contour of a phrase. **Weak form** — the reduced pronunciation of a function word.

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Final note: the ranked shortlist

This is the only ranking in the document, and it is the last thing I'll say.

In order:

1. **Make weak syllables genuinely weak.** (2.3, 3.10, 4.2) Because it's the cue English listeners weight most heavily for stress (3.2), and because unreduced syllables plant false word boundaries in your listener's parse (1.3). Everything else is downstream of this.
2. **Audit and fix the word stress of your professional vocabulary.** (3.9, 3.12) Fastest large gain available. It's knowledge, so it's fast, and misplaced stress is the most destructive single error category (Field 2005).
3. **Put the nucleus on the word that matters, and deaccent what's given.** (6.2, 6.3, 6.6) The difference between being decoded and being understood, and it moves how competent you seem (Hahn 2004).
4. **Attach the pitch movement to the stressed syllable, not the phrase edge.** (7.1, G3) The single fix that most reduces the "French sing-song" impression.
5. **Learn the fall-rise.** (7.3 tune 4, G6) The largest gain in how you're *received* rather than how well you're understood.

6. **Train your perception, with talker variability, immediate feedback, and a score you write down.** (9.2) Because of 1.6 this is the constraint on everything above, and §9.2 gives you a protocol with effect sizes attached rather than a vague instruction to listen more.

Six things. Everything else in this document is detail, support, or explanation of why those six matter.

And one thing that is not on the list but should be said: your stated target — completely clear, prosodically native-like, still audibly French — is both achievable and, in my view, better than the alternative. Aim at it deliberately rather than treating it as a compromise.